BSc Thesis — Master Methodology & Reference Document

ML-Driven Inverse Design of Microring Resonators for Co-Packaged Optics / High-Bandwidth Optical Interconnects

Unified Single Source of Truth — Version 1.3

Supersedes: Thesis Methodology & Reference Plan v1.0, Thesis Supporting Chapters & Reference Appendix v1.0, Errata & Patch v1.1 (merged into the single v1.1 master document), v1.2, and now this v1.3 revision.

v1.3 changes: applies sixteen corrections from a third, fully independent audit in which every numbered equation was re-derived and every numeric claim re-checked against the primary papers rather than this document's own citations. Corrections are prose-, attribution-, definition- and framing-level only: *no fixed default, equation, threshold, phase or objective was removed or weakened*, and none was added. One new section (Section 15, Future Scope Beyond This Thesis) was added and is explicitly outside this thesis's scope. See the Version History for the full v1.0 → v1.3 changelog.

No separate errata file is needed — this document is the complete, standalone reference.

Prepared for: BSc final-year research paper, targeting MSc/PhD applications (Netherlands and international)

Prerequisite skills assumed: Python, foundational ML/DL, learning photonics simulation

Primary simulation tool: Ansys Lumerical FDTD (3D) with Python API (`lumapi`) — with an open-source contingency path, see Section 7[v1.1]

Comparison methods: Random Forest, XGBoost, Deep Neural Network, Tandem Network

How to use this document: every phase specifies exact parameter values, equations, hyperparameter grids, and pass/fail thresholds. Where a choice must be made, a specific default is fixed so that no step requires an undocumented judgment call. Adjust only if your own convergence tests or compute budget require it, and log any deviation in Section 7 (Risk/Limitation Log). Passages marked [v1.1] [v1.2] or [v1.3] were corrected or added relative to earlier versions; see the Version History at the end for the full changelog.

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Part I — Methodology & Reference Plan

1 Title, Abstract, and Research Problem Statement

1.1 Working Title

ML-Driven Inverse Design of Microring Resonators for Co-Packaged Optics / High-Bandwidth Optical Interconnects

1.2 Abstract (Draft — to be refined after results are in)

Co-packaged optics (CPO) and other high-bandwidth optical interconnect architectures increasingly rely on silicon-photonic microring resonators (MRRs) as compact, wavelength-selective filtering elements for dense wavelength-division multiplexing (WDM). Designing an MRR that simultaneously meets a target resonance wavelength, free spectral range (FSR), quality factor (Q), and insertion loss (IL) requires either slow full-wave electromagnetic simulation sweeps or iterative optimization, both of which scale poorly as the design space grows. Machine-learning (ML)-based surrogate modelling and inverse design have been demonstrated for microring resonators, but prior work has addressed these methods in isolation and in application contexts (frequency-comb dispersion engineering, quantum squeezed-light generation, biosensing, generic channel-dropping filters) that do not explicitly target CPO/interconnect specifications, and none of it directly compares classical ensemble machine learning (Random Forest, XGBoost) against deep neural networks (DNNs) and tandem networks on identical data under identical metrics. This thesis builds an open, reproducible Lumerical FDTD dataset of add-drop MRR geometries (ring radius, waveguide width, coupling gap, coupling length) labelled with their resonance wavelength, FSR, loaded Q-factor, and insertion loss, then trains and rigorously benchmarks Random Forest, XGBoost, a feed-forward DNN, and a tandem network for both forward (geometry → response) and inverse (response → geometry) prediction. Model outputs are further translated into CPO-relevant design implications — channel count vs. FSR, per-ring insertion-loss budget across a cascaded filter bank, and 3-dB bandwidth vs. target data rate — rather than reported as abstract regression accuracy alone. Results are expected to clarify (i) which method family is most sample-efficient in the small-dataset regime realistic for a full 3-D FDTD study, (ii) whether the well-documented one-to-many inverse-design ambiguity manifests for this specific four-parameter MRR design space, and (iii) what prediction-error tolerance is acceptable before it threatens a realistic CPO link budget.

[v1.1] addition, Fix 5: **This is a controlled comparative benchmarking study, not a new inverse-design algorithm**: Random Forest, XGBoost, the DNN, and the tandem network are all established techniques from the literature (Papers 1–12), applied here for the first time under identical data and identical metrics to this specific device and application context.

[v1.2] addition, Fix 4a: Of the seven objectives below, **Objective O6** — whether the one-to-many inverse-design ambiguity documented in Paper 3 actually manifests for this specific microring design space, given that Papers 7 and 9 report the opposite in their own, differently-bounded spaces — is this thesis's **primary citable scientific contribution**: a genuine, currently-open tension in the literature that this thesis resolves empirically, independent of which way the result falls. The remaining objectives (O1–O5, O7) establish the controlled, rigorous comparison that makes that resolution credible, rather than being separate claims of comparable weight.

[v1.3] addition, Fix 14a — **what makes this design space the sharpest available test of O6.** The three papers in tension differ less in the size of their design spaces than in *how much information their inverse networks receive about the target response*. Liu et al. [3] input a 200-point spectrum and recover 20 layer thicknesses (training stalls); Chen & Jiang [7] input a 100-point spectrum and recover 5 parameters (converges); Adibnia et al. [9] input an 800-point spectrum and recover 2 parameters (converges, with acknowledged geometry mismatch). This thesis inputs *four scalars* and recovers four parameters — an order of magnitude less response information than the smallest prior case. Near-duplicate responses are therefore substantially more likely here than in any published precedent, which makes this the most discriminating test of Liu et al.'s hypothesis yet attempted, and makes either outcome informative.

1.3 Research Problem Statement

High-bandwidth optical interconnects for co-packaged optics require microring-resonator filters whose four core specifications — resonance wavelength (λ_{res}), free spectral range (FSR), loaded quality factor (Q_L), and insertion loss (IL) — must be hit simultaneously and reliably across a manufacturable design space. The conventional design loop (parameter sweep in a full-wave electromagnetic solver, inspect the spectrum, adjust, repeat) is accurate but slow: a single 3-D FDTD run of a resonant structure can take minutes to tens of minutes, and a systematic sweep over four coupled geometric parameters multiplies that cost combinatorially. Data-driven inverse design — training a model to map a desired optical response directly back to a candidate geometry — promises to collapse this loop into a single fast query, but the literature that motivates this promise is scattered across at least three disjoint threads: (a) classical machine learning applied to isolated (non-bus-coupled) ring dispersion, (b) deep neural networks applied to bus-coupled filters or plasmonic switches with no classical-ML comparison, and (c) the most recent physics-informed graph-neural-network approaches applied to biosensing rather than data communication. No existing study performs a controlled, single-dataset, same-metric comparison of classical ensemble ML, a plain DNN, and a tandem network for a bus-coupled add-drop MRR filter, and none frames its evaluation explicitly around CPO/interconnect performance implications (channel count, cascaded-filter power budget, 3-dB bandwidth vs. data rate). This thesis addresses that specific, well-defined problem.

2 Research Gap Summary

2.1 What Exists (synthesized across all 12 reviewed papers)

- **Classical ML for microresonator inverse design** has been demonstrated exactly once in the reviewed literature (Pal et al., Paper 6): Random Forest beats Decision Tree (MAPE 3–15% vs. 9–27%) for predicting an isolated ring's radius/width/height from its dispersion profile. There is no bus waveguide and no coupling gap in that parameter space at all.
- **DNN forward+inverse modelling of bus-coupled MRR filters** has been demonstrated (Chen & Jiang, Paper 7, on a 4-ring channel-dropping filter via 2-D Meep FDTD, 25,000 samples) and for a structurally analogous but physically different plasmonic nonlinear switch (Adibnia et al., Paper 9, via FDTD, 147,456 samples). Both use a single, non-tandem, direct

inverse network and predict the full transmission spectrum, not scalar CPO specs.

- **The tandem-network solution to the one-to-many inverse-design problem** is well-established in the broader nanophotonics literature (Liu et al., Paper 3, the originating work; Wiecha et al., Paper 1, and Khaireh-Walieh et al., Paper 2, the two review/tutorial sources) but has never been applied to a microring resonator specifically, let alone compared against classical ML on the same problem.
- **A dedicated, openly released MRR geometry-vs-optical-response dataset** exists (Valencia-Garzón et al., Paper 10; 4,719 samples, FIMMWAVE/EME) but covers only an all-pass (single-bus, no coupling length) topology and performs no ML modelling at all — its own User Notes section explicitly recommends regression, random forest, and neural-network inverse design as future work.
- **The most recent and most sophisticated approach found** (Torabi et al., Paper 12, 2026) uses a physics-informed graph neural network on a four-parameter MRR space, targeting biosensing (refractive-index sensitivity) rather than data communication, and — like every other paper reviewed — includes no classical-ML baseline. [v1.3] Fix 11: *the earlier characterisation of Paper 12's design space as "almost identical" to this thesis's own was an overstatement and has been corrected.* Paper 12 varies (R, w, h, g) over 0–500 μm, 1.5–3.0 μm, 1.5–3.0 μm and 0.1–0.3 μm respectively — a large polymer-scale biosensor ring, not an SOI telecom filter. Three of the four parameters share a *name* with this thesis's own; the fourth differs, and it differs in exactly the respect this thesis claims as structural novelty (Paper 12 varies waveguide height h and holds coupling length fixed; this thesis fixes $h = 220$ nm and varies L_c). The numerical ranges differ by one to two orders of magnitude. The correct statement is that Paper 12 is the *closest four-parameter microring inverse-design study by dimensionality and recency*, not by design space.
- Two further papers use ML/optimization on microring resonators for applications outside the classical-ML-vs-DNN scope entirely: Bayesian Optimization with a Gaussian-Process surrogate for quantum squeezed-light generation (Dizaji & Habibiyan, Paper 8), and microring resonators used as computing hardware to accelerate someone else's neural network (Tang et al., Paper 11) — the inverse relationship to this thesis's topic, useful only for scoping and realistic Q/extinction-ratio calibration numbers.

2.2 What Is Missing

1. No study performs a **controlled, single-dataset, same-metric comparison** of Random Forest, XGBoost, a plain DNN, and a tandem network for microring resonator inverse design.
2. No study models a **bus-coupled, add-drop MRR with an explicit coupling-length parameter** (as opposed to a point-coupler model) while simultaneously predicting insertion loss as a first-class scalar target.
3. No study frames its four target quantities explicitly as **CPO/interconnect design specs** (channel count via FSR, cascaded power budget via IL, filter bandwidth vs. data rate via Q/FWHM) rather than generic photonics regression accuracy.
4. No study explicitly tests, for a microring resonator specifically, whether the **one-to-many/non-uniqueness problem** that motivates tandem architectures in the broader nanophotonics literature (Paper 3) actually manifests at the scale of a compact, physically-bounded four-parameter MRR design space — an open, unresolved tension visible in the literature itself (Paper 3 predicts failure of naive inverse networks; Papers 7 and 9 report naive inverse networks working acceptably well in their own, differently-bounded design spaces). [v1.3] Fix 14b: note that Papers 7 and 9 judged "working acceptably" in *response* space after re-simulation, not in geometry space — Adibnia et al. state plainly that their predicted and actual structures are distinct while their spectra are not. Objective O6 is therefore stated against both criteria (Section 3).

5. No study reports **sample-efficiency curves** (accuracy vs. training-set size) for this specific comparison, despite the field's own repeated, independent acknowledgement (Papers 2, 4, 5, 7, 9) that training-data coverage and cost are the dominant practical bottleneck.

2.3 Why This Matters Specifically for Co-Packaged Optics

CPO and other tightly-integrated optical interconnect architectures deliberately operate microring resonators in a **moderate-Q regime** ($\sim 10^3$–10^5), because bandwidth — not spectral purity — is the primary design driver: a ring narrow enough to reach the ultra-high Q values pursued for frequency-comb generation (Paper 6, measured $Q_L \approx 5 \times 10^5$) or squeezed-light sources (Paper 8, $Q_L = 1.6$–2.8×10^5) would be too narrowband to pass a modulated data signal and too sensitive to thermal/fabrication drift for a production interconnect. This is a materially different design target from every microring-specific paper reviewed, and it is why insertion loss, coupling-length-tunable coupling strength, and FSR (as a proxy for usable WDM channel count) — rather than Q maximization alone — must be treated as co-equal, simultaneously-optimized objectives.

[v1.3] Fix 8 — **scope note on the upper end of the moderate-Q band.** The $\sim 10^3$–10^5 band above describes the regime CPO *targets*; it is not yet a verified claim about what the Phase 1.2 design space will actually *populate*. Inverting Eq. 6 at $R = 15\,\mu$m, $n_g \approx 4.2$, $L_c = 0$ and $\sim$2 dB/cm propagation loss, $Q_L = 10^5$ requires $t_1 t_2 a \approx 0.992$, i.e. a power coupling of roughly 0.6% per coupler; an analytic evanescent-decay estimate for a 450×220 nm SOI TE coupler at the $g = 350$ nm upper bound gives several times more coupling than that. The realistically populated band may therefore be closer to 10^3–10^4. This estimate is approximate and is *not* a basis for changing the design space: the Phase 1.6 coupling-boundary sub-study (item 4) exists precisely to settle it empirically. Treat 10^3–10^5 as the target band and the Phase 1.6 result as the authority. Nothing downstream depends on the upper end — both Phase 7 CPO scenarios require only $Q_L \approx$ 1,500–8,000.

[v1.1] replacement, Fix 5: This specific combination — classical-ML-vs-DNN-vs-tandem comparison, framed around this moderate-Q, insertion-loss-aware, CPO-relevant design regime — was not identified anywhere in the twelve papers reviewed for this thesis (Section 2.1); this is **evidence of a gap in the literature surveyed, not a claim of priority across the field at large**.

3 Research Objectives

Objectives are numbered and stated with an explicit, measurable pass/fail threshold. All thresholds are defaults grounded in the metrics reported by the closest comparable papers (chiefly Papers 6, 7, 9, and 12); revise only if Phase 1 data quality forces a change, and log any revision in Section 7.

[v1.2] addition, Fix 4a: **Which objective is the primary contribution.** Objective O6 below is this thesis's central, citable scientific finding, not one of seven equally-weighted checkboxes. The literature reviewed contains a genuine, unresolved tension: Liu et al. [3] (H_A) show that a naive, non-tandem inverse network can fail to converge on a sufficiently large/redundant design space, while Chen & Jiang [7] and Adibnia et al. [9] (H_B) report naive inverse networks converging acceptably in their own, differently-bounded design spaces. This thesis is positioned to report, for the first time, which of H_A or H_B holds for a compact, physically-bounded, four-parameter add-drop microring space — and **either outcome is a genuine, reportable finding**: a clean resolution in either direction is more scientifically interesting than an ambiguous result, and neither outcome should be treated as a failure of the thesis. Objectives O1–O5 and O7 exist to make that resolution methodologically credible (a controlled, adequately-powered, fairly-benchmarked comparison), not to stand as independent contributions of equal weight to O6.

O1. Dataset construction.
Generate and publicly document a Lumerical FDTD dataset of ≥ 800 add-drop microring resonator geometries (target 1,000; stretch target 1,500), each labelled with λ_{res}, FSR, Q_L, and IL, and validate the simulation setup against at least one independently published, experimentally characterized MRR to within $\leq 10\%$ relative error on FSR (matching the 2.6–9% precedent set by Paper 10).

O2. Forward surrogate accuracy.
Achieve, on the held-out test set, $R^2 \geq 0.90$ and MAPE $\leq 10\%$ for at least three of the four target quantities using the best-performing forward model, with all four quantities individually reported (no target may be omitted from reporting even if it under-performs).

O3. Inverse design accuracy.
Achieve, for at least one comparison method, an aggregate RMS Relative Geometric Error (RMS-RGE, Section 4 Phase 6) $\leq 15\%$ against held-out target specs, and Root-Mean-Square Closed-Loop Response Error (re-simulating the predicted geometry in Lumerical) $\leq 20\%$ against the originally requested target response. [v1.3] Fix 4: RMS-RGE is expressed as a percentage (Eq. 15), and the L_c term uses the range-normalised convention defined in Phase 6.2, which is required because $L_c = 0$ is a valid design point and a target-normalised relative error is undefined there.

O4. Controlled four-way comparison.
Report Random Forest, XGBoost, DNN, and Tandem Network results on an identical train/-validation/test split with identical metrics (MAPE, RMSE, R^2 per target; RMS-RGE and closed-loop error for inverse design), with no method excluded from any metric regardless of relative performance.

O5. Sample-efficiency characterization.
Produce an accuracy-vs.-training-set-size curve for all four methods (at ≥ 5 training-set sizes) and explicitly state which method is most accurate in the ≤ 300-sample regime and which is most accurate at full dataset size — directly answering the small-data-regime question motivating the inclusion of classical ML.

O6. One-to-many ambiguity test.
Explicitly report whether the naive (non-tandem) direct inverse DNN converges acceptably or exhibits the non-convergence/high-error pattern documented in Paper 3, thereby empirically resolving — for this specific MRR design space — the open tension between Paper 3 and Papers 7/9 identified in the literature review. [v1.3] Fix 14c — **dual reporting**

criterion (clarification, not an added requirement). "Converges acceptably" must be reported against *both* of the following, because the two sides of the tension used different criteria:

- **Geometry-space criterion:** $R^2 \geq 0.80$ against ground-truth geometry (the original O6 threshold, unchanged);
- **Response-space criterion:** closed-loop re-simulated response error, which is what Chen & Jiang [7] and Adibnia et al. [9] actually used when reporting success.

Both numbers are already produced by Phase 4.4 (step 1 and step 4 respectively); this clarification mandates that both are *stated* when O6 is claimed, and that any divergence between them is discussed. A result in which geometry-space R^2 fails while response-space error passes is precisely Adibnia et al.'s reported situation and is a legitimate, publishable O6 outcome — not a null result.

O7. CPO application-relevance translation.
For at least two worked CPO scenarios (Section 4, Phase 7), map each comparison method's inverse-design output to a concrete channel-count, cascaded insertion-loss-budget, and 3-dB-bandwidth-vs.-data-rate outcome, and state explicitly which method(s) meet the scenario's specification within its stated tolerance.

4 Methodology

Seven sequential phases. Each phase states its purpose, exact tools, step-by-step actions, governing equations, and expected output. Phases 1–2 must be completed before any modelling begins; Phases 3–5 can proceed in parallel once Phase 2 is complete; Phase 6 requires all of Phases 3–5; Phase 7 requires Phase 6.

4.1 Phase 1 — Simulation Dataset Generation

Purpose: Produce a labelled dataset mapping add-drop microring resonator geometry to its optical response, of sufficient size and coverage to train and fairly compare four ML method families, at a scale realistically achievable within a BSc compute/time budget.

1.1 Device Definition (fixed, non-ambiguous)

- **Topology:** Add-drop (two-bus, four-port) racetrack microring resonator — through port, drop port, input port, add port. Racetrack (straight coupling section) rather than pure circular coupling, so that coupling length is a genuine, continuously tunable geometric parameter rather than a fixed point-coupling interaction.
- **Platform:** Silicon-on-insulator (SOI). Core: silicon, fixed waveguide height/thickness $h = 220\,\text{nm}$ (standard SOI device layer). Cladding: SiO_2 (upper and lower). Fundamental TE mode only.
- **Operating band:** C-band, wavelength scan window 1500–1600 nm, nominal target wavelength 1550 nm.
- **Material dispersion:** Use Lumerical's built-in dispersive material models for Si and SiO_2 (or the Palik dataset if built-in models are unavailable in your license tier); do not use a constant, non-dispersive refractive index.

1.2 Design Parameters and Ranges (fixed, non-ambiguous)

Four free geometric parameters, sampled continuously (not on a fixed grid) via Latin Hypercube Sampling (LHS) across the ranges below. Both couplers (input–through and ring–drop) use

the same gap and coupling length for symmetry and to keep the design space at exactly four dimensions; asymmetric couplers are a valid but explicitly out-of-scope extension (Section 7).

Symbol	Parameter	Range	Rationale
R	Ring radius	5–15 μm	Spans FSR $\approx$ 6–18 nm at the nominal n_g fixed below; small enough to keep footprint CPO-relevant (cf. Paper 7's $R \approx 5\,\mu$m footprint precedent).[v1.3]
w	Waveguide width	400–500 nm	Standard single-mode strip width on 220 nm SOI at 1550 nm (cf. Papers 4, 7, 11).
g	Coupling gap	150–350 nm	Spans near-critical to weakly-coupled regimes; the upper bound follows the bound-derivation *method* of Papers 7 and 8, with the numeric value to be confirmed by the Phase 1.6 coupling-boundary sub-study.[v1.3]
L_c	Coupling length	0–3 μm	$L_c = 0$ recovers a pure point/pulley coupler; up to 3 μm gives continuously tunable coupling strength independent of gap, the parameter missing from Papers 6, 10, and 12.

[v1.3] Fix 7 — **nominal group index, now stated explicitly.** The FSR range claimed for R above cannot be checked unless n_g is fixed, and it was not stated in v1.0–v1.2. **Fixed nominal value for all design-space reasoning: $n_g \approx 4.2$ for the fundamental TE mode of a $\sim$450$\times$220 nm SOI strip waveguide at 1550 nm.** This is corroborated by Paper 11, which reports a fabricated 440$\times$220 nm SOI TE ring of $R = 20\,\mu$m with a measured FSR of 4.4 nm, implying $n_g = 4.35$. Verification at $n_g = 4.2$, $L_c = 0$: $R = 5\,\mu$m $\Rightarrow$ FSR = 18.2 nm; $R = 15\,\mu$m $\Rightarrow$ FSR = 6.07 nm — the claimed 6–18 nm span is correct. Across the full four-dimensional space (L_c up to 3 μm, which lengthens L and shortens FSR) the true span is $\approx$ 5.7–18.2 nm.

Two attribution corrections follow. (i) Paper 7's $R \approx 5\,\mu$m/FSR $\approx$ 21 nm pair implies $n_g \approx 3.6$, because Paper 7 is a *2-D, TM-polarised, 0.3 μm-wide silicon* structure — its FSR figures are cited here as a *footprint* precedent only and do not transfer numerically to 220 nm SOI TE. (ii) The "coupling strength decays to ≈ 0 by $\sim$300–350 nm for this waveguide cross-section" parenthetical previously attached to the g row traced only to Paper 7 (same different cross-section), and Paper 8 bounds the coupling *coefficient* $\kappa \leq 0.5$ directly rather than bounding the gap at all. The g range is unchanged; what has changed is that its numeric justification is now correctly stated as *pending the Phase 1.6 sub-study 4 measurement* rather than as established by Papers 7 and 8.

Design-space rationale: This range is deliberately chosen to populate the moderate-Q regime ($\sim 10^3$–10^5) relevant to data-communication filters, in contrast to the ultra-high-Q regime targeted by Papers 6 and 8 for comb generation and squeezed light. **Confirm this empirically in Phase 1.6 before committing to the full run** — and see the Section 2.3 scope note[v1.3] on the upper end of this band.

1.3 Governing Equations

Round-trip phase accumulated in the ring:

$$\phi = \beta L, \qquad \beta = \frac{2\pi n_{eff}}{\lambda}, \qquad L = 2\pi R + 2L_c \tag{1}$$

where n_{eff} is the waveguide's effective index (obtained from the FDTD mode solver, not assumed) and L is the total round-trip length including the two straight coupling sections.

Add-drop ring transmission (self-coupling coefficients t_1, t_2 for the two couplers, round-trip amplitude transmission a accounting for propagation loss):

$$T_{through} = \frac{t_2^2 a^2 - 2t_1 t_2 a\cos\phi + t_1^2}{1 - 2t_1 t_2 a\cos\phi + (t_1 t_2 a)^2} \tag{2}$$

$$T_{drop} = \frac{(1-t_1^2)(1-t_2^2)\,a}{1 - 2t_1 t_2 a\cos\phi + (t_1 t_2 a)^2} \tag{3}$$

with the lossless-coupler power relation $\kappa^2 + t^2 = 1$ linking self-coupling t to cross-coupling (coupling coefficient) κ at each coupler.

Free spectral range:

$$\mathrm{FSR} = \frac{\lambda_{res}^2}{n_g L} \tag{4}$$

where n_g is the group index (extract numerically from the FDTD dispersion data; do not assume $n_g \approx n_{eff}$).

Loaded quality factor (two equivalent forms — use the FWHM form to extract Q_L directly from simulated spectra, and the analytical CMT form for physics-informed bounds-setting in Phase 1.6 and the physics-informed loss extension in Phase 4):

$$Q_L = \frac{\lambda_{res}}{\mathrm{FWHM}} \qquad \text{(from simulated spectrum)} \tag{5}$$

[v1.1] **Fix 1 — corrected analytical Q_L formula:** Was missing a factor of π and silently dropped L_c. Re-derived from the standard finesse relation $F = \pi\sqrt{t_1 t_2 a}/(1 - t_1 t_2 a)$ and $Q_L = f_{res} \cdot F/\mathrm{FSR}$ (with $\mathrm{FSR}_f = c/(n_g L)$):

$$Q_L = \frac{\pi\, n_g L \sqrt{t_1 t_2 a}}{(1 - t_1 t_2 a)\,\lambda_{res}}, \qquad L = 2\pi R + 2L_c \text{ (as in Eq. 1)} \tag{6}$$

For a pure point-coupled ring ($L_c = 0$), this reduces to $Q_L = 2\pi^2 n_g R\sqrt{t_1 t_2 a}/[(1 - t_1 t_2 a)\lambda_{res}]$. Cross-checked against Paper 8's own Eq. (6) (Sadeghi Dizaji & Habibiyan, *Sci. Rep.* 15:372, 2025, p. 5), which states $Q_L = 2\pi^2 n_g R\sqrt{ta}/[(1 - ta)\lambda_{res}]$ for their single-coupler geometry (attributed there to Bogaerts et al. 2012 [13]). Not independently verified against the typeset equation in Bogaerts et al. 2012 itself (paywalled); treat as strongly triangulated but confirm directly if you obtain journal access.

[v1.3] **Fix 2 — correction to the substitution described above (the equation itself is unchanged and confirmed).** Earlier versions described the single-coupler-to-two-coupler generalization as $t \to \sqrt{t_1 t_2}$. That is wrong, and it is inconsistent with Eqs. 2–3, which use the *product* $t_1 t_2$ throughout. The correct substitution is

$$t \;\longrightarrow\; t_1 t_2,$$

i.e. the round-trip field survival factor for two couplers is $t_1 t_2 a$, exactly as it appears in the denominators of Eqs. 2–3. Substituting $t = t_1 t_2$ into Paper 8's Eq. (6) reproduces the $L_c = 0$ reduction above *exactly*; substituting $t = \sqrt{t_1 t_2}$ would instead give $\sqrt{\sqrt{t_1 t_2}\,a}\,/\,(1 - \sqrt{t_1 t_2}\,a)$, which is not Eq. 6. Eq. 6 was independently re-derived from the finesse relation in the v1.3 audit and is **confirmed correct**; only the prose describing its provenance was in error.

Insertion loss (drop-port peak transmission expressed in dB):

$$\mathrm{IL}\,[\mathrm{dB}] = -10\log_{10}\left(T_{drop,\,peak}\right) \tag{7}$$

1.4 Simulation Tool Setup

1. **Solver:** Ansys Lumerical FDTD (3-D), driven programmatically via the Python API (`lumapi`) — not the GUI — so that all 800–1,500 runs are scripted and reproducible. [v1.1] If Lumerical access is unavailable, see the MEEP/Tidy3D contingency in Section 7 and the tool-substitution notes below.
2. **Source:** Mode source injecting the fundamental TE mode at the input port; broadband pulse spanning 1500–1600 nm so at least two adjacent resonance dips fall inside the scan window (required to measure FSR directly rather than only estimate it).
3. **Monitors:** Frequency-domain power monitors at the through port and the drop port, recording $T_{through}(\lambda)$ and $T_{drop}(\lambda)$ at a spectral resolution fine enough to resolve FWHM for the highest-Q samples in the range (minimum 2,000 frequency points across the scan window; verify against the narrowest expected linewidth in Phase 1.6).
4. **Boundary conditions:** Perfectly matched layers (PML) on all simulation-domain boundaries.
5. **Mesh:** Non-uniform auto mesh globally, with an explicit mesh-override region around each coupling gap (finest mesh where coupling physics is most sensitive, following the coupling-region mesh sensitivity documented in Paper 10). Determine the override resolution via the convergence study in Phase 1.6, not by default settings.

[v1.1] **Fix 2 — what changes if the MEEP/Tidy3D fallback (Section 7) is triggered:**

- **Solver/scripting (item 1):** `lumapi` → MEEP's native Python interface (`import meep as mp`); geometry defined via `mp.Cylinder`/`mp.Block` objects rather than `lumapi` geometry primitives (also changes Phase 1.5 step 1).
- **Monitors (item 3):** Lumerical frequency-domain power monitors → MEEP DFT flux monitors (`mp.FluxRegion`/`add_flux`). The $\geq$ 2,000-point spectral-resolution requirement carries over unchanged.
- **Mesh (item 5):** MEEP has no override-region concept; it uses a single global resolution parameter (pixels per unit length) plus subpixel averaging. The Phase 1.6 mesh-convergence study must be re-run against resolution, not an override-region size, if this fallback triggers.
- **Material dispersion:** Lumerical's built-in models → MEEP's material library, or a manually-fit Sellmeier/Lorentzian susceptibility model built from the same Palik n, k data already named as the Lumerical fallback — no new data source needed, only a different way of feeding it to the solver.
- **Boundary conditions (item 4):** Unchanged — MEEP also uses PML (`mp.PML`).
- **2-D vs. 3-D:** Chen & Jiang's precedent (Paper 7) is 2-D MEEP. Default: stay in 3-D MEEP to preserve this thesis's stated fidelity differentiator; treat 2-D MEEP as a further, nested fallback only if the timing gate in Fix 3 below is also triggered.

1.5 Step-by-Step Data Generation Procedure

1. Build one parameterized Lumerical simulation template (script-based, using `lumapi` geometry primitives) with R, w, g, L_c exposed as script variables.
2. Run the mesh-convergence study (Phase 1.6) once, on a single representative geometry, to fix mesh settings for all subsequent runs.
3. Validate the template against ≥ 1 independently published, experimentally characterized MRR (Phase 1.6) before mass production.

4. Generate the full parameter sample list via Latin Hypercube Sampling across the four ranges in Section 1.2 (fixed random seed = 42 for reproducibility). Target N = 1,000 samples (minimum acceptable N = 800; stretch target N = 1,500 if compute time allows — see Section 8, Timeline).
5. Batch-execute: for each sample, instantiate the geometry, run the FDTD simulation, extract $T_{through}(\lambda)$ and $T_{drop}(\lambda)$, and write raw spectra plus derived quantities to a structured output file. Run unattended in batches (overnight/queued) given the multi-minute-to-tens-of-minutes per-sample cost typical of 3-D FDTD (cf. Papers 4 and 9 for realistic per-sample and total compute-cost benchmarks, and the timing-calibration gate in Phase 1.6 below[v1.1]).
6. Post-process each spectrum (automated script, not manual inspection) to extract: λ_{res} (drop-port peak nearest 1550 nm), FSR (spacing to the adjacent drop-port peak), FWHM (from the drop-port peak used for Q_L), Q_L (Eq. 5), and IL (Eq. 7). Also record n_{eff} and n_g extracted from the mode solver for later use in the analytical formulas (Eqs. 4, 6) and the Phase 4 physics-informed loss term.
7. Flag and log (do not silently discard) any sample where no resonance dip is found in the scan window, where the extracted FWHM is unresolved by the monitor's spectral resolution, or where $T_{drop} > 1$ (a numerical/rounding artefact documented in Paper 7) — these become explicit data-quality-log entries handled in Phase 2.1.
8. [v1.2] **Publish the dataset.** Push the cleaned dataset (CSV/Parquet), the data-quality log, the timing-calibration record, the mesh-convergence report, and the literature-validation report to a public GitHub repository, before or alongside thesis submission — this is the concrete action that fulfils Objective O1's "publicly document" requirement (see Section 6, Git/GitHub row), not merely a stated intention in the tools list.

1.6 Required Sub-Studies (do not skip)

[v1.1] Fix 3 — new sub-study, run first, before the original three (renumbered below to 2–4).

1. **Timing calibration**[v1.1] (run first, before committing to mass production). Run one complete 3-D FDTD sample at the mesh resolution and spectral point count fixed in sub-study 2 below, and record actual wall-clock time per sample.
   Decision rule, calibrated against the Section 8 execution window (originally 4 weeks, Weeks 3–6): assume ∼8 reliable unattended-batch hours/day on typical BSc-accessible hardware (a shared lab machine or personal workstation, not a dedicated cluster). Over 28 calendar days this gives $8 \times 28 = 224$ hours = 13,440 minutes of usable compute, i.e. a break-even of ≈ 13.4 min/sample for N=1,000 and ≈ 16.8 min/sample for the N=800 floor (Objective O1). **Threshold: 15 minutes/sample**, sitting inside this break-even band.
   - $\leq$ 10 min/sample: full stretch target N=1,500 remains realistic.
   - 10–15 min/sample: target N=1,000 remains realistic; drop the stretch target.
   - 15–25 min/sample: reduce target to the N=800 floor (already the Objective O1 minimum); document as a scope adjustment, not a failure.
   - $>$ 25 min/sample: even N=800 will not fit the window — trigger the 2-D FDTD downgrade (Section 7).
2. **Mesh convergence study:** sweep mesh resolution on one representative geometry; confirm the transmittance gradient falls below $\sim 10^{-4}$ before mass production (methodology directly modelled on Paper 10, Section 2.3).
3. **Literature validation:** reproduce the reported FSR (and, if available, Q or IL) of at least one independently published, experimentally characterized SOI or Si_3N_4 MRR using your exact simulation template; report percentage error (target $\leq$ 10%, matching the 2.6–9% precedent

in Paper 10).

4. **Coupling-strength boundary check:** for the extremes of the g/L_c range, confirm via an auxiliary coupling-region simulation that the resulting coupling coefficient spans from near-zero to near-critical coupling (methodology directly modelled on Papers 7 and 8), confirming the moderate-Q design-space rationale in Section 1.2. [v1.3] Fix 8: this sub-study is the *sole authority* on (a) whether $g = 350\,\text{nm}$ genuinely reaches the weak-coupling limit for a $450 \times 220\,\text{nm}$ SOI TE cross-section, and (b) what Q_L ceiling the design space actually reaches. Record the measured $\kappa^2(g, L_c)$ at the range extremes and the implied Q_L ceiling explicitly in the Phase 1 output, and reconcile it against the 10^3–10^5 target band in the thesis text either way.

Expected output of Phase 1: A structured dataset file (CSV/Parquet) of $N \geq 800$ rows, columns $= [R, w, g, L_c, n_{eff}, n_g, \lambda_{res}, \text{FSR}, \text{FWHM}, Q_L, \text{IL}]$, plus: (a) a data-quality log of flagged/excluded samples, (b) a timing-calibration record and mesh-convergence report[v1.1], (c) a literature-validation report with quantified percentage error, (d) [v1.2] the public GitHub repository containing all of the above, and (e) [v1.3] the measured coupling-boundary record from Phase 1.6 sub-study 4.

4.2 Phase 2 — Data Preprocessing

Purpose: Turn the raw Phase 1 dataset into clean, split, and appropriately-scaled train/validation/test sets, applied identically to all four comparison methods so that Phase 6's cross-method comparison is fair.

2.1 Data Cleaning

1. Remove or explicitly re-simulate (do not silently impute) any sample flagged in Phase 1.5 step 7 (no resonance found, unresolved FWHM, or $T_{drop} > 1$ artefact).
2. Clip any residual T_{drop} or $T_{through}$ values fractionally above 1.0 (floating-point/flux-normalization artefacts, cf. Paper 7) to exactly 1.0 before computing IL.
3. Record the final clean sample count N_{clean}; this is the number used in Objective O1.

2.2 Exploratory Data Analysis and Coverage Check

1. Plot the marginal distribution of each of the four outputs (λ_{res}, FSR, Q_L, IL) and each of the four inputs.
2. Plot pairwise scatter plots of each input against each output to visually confirm the expected monotonic trends (e.g. FSR decreasing with R, Q_L increasing with g).
3. Explicitly identify any under-sampled region of the output space (e.g. very high-Q_L tail) — this is the single most commonly self-reported failure mode across the reviewed literature (Papers 2, 4, 5, 9). If a clear gap exists, generate a small supplementary batch of LHS samples targeted at that region before finalizing the split.

2.3 Feature and Target Transformation

- **Inputs (R, w, g, L_c):** Min–Max normalize to $[0, 1]$ for the DNN and Tandem Network. For Random Forest and XGBoost, train and evaluate both on normalized and on raw (unnormalized) inputs, and report both — Paper 6 found normalization can degrade tree-ensemble accuracy on some targets, so this must be tested empirically, not assumed.

  [v1.3] Fix 13 — **note on how Paper 6 is being read here.** Paper 6's prose states flatly that normalization "deteriorates the performance of the RF model". Its own Table 1 shows a *mixed* result: radius MAPE worsens (12.52 → 21.43) while width (3.14 → 1.89) and height

(15.01 → 6.34) both improve substantially. The per-target reading used in this thesis follows Paper 6's *data*, not its summary sentence. State this explicitly in the thesis text so a reader checking Paper 6's abstract does not conclude the source has been misquoted.

- **Target — Q_L:** apply $\log_{10}$ transform before training ($y = \log_{10} Q_L$), since Q_L realistically spans several decades in this design space (target band 10^3–10^5; see the Section 2.3 scope note[v1.3] and confirm the achieved span from Phase 1.6 sub-study 4). A linear-scale MSE loss would be dominated by the high-Q_L tail (cf. Paper 9's MSE-vs-MSLE finding; note that Paper 9 itself describes the MSE/MSLE difference as slight and selected MSLE as the lower-loss option among hyperparameters[v1.3]). Report all Q_L error metrics after transforming predictions back to linear scale. **The $\log_{10}$ transform remains the fixed default regardless of the achieved span**, because a multi-decade spread is expected on any of the candidate bands.
- **Targets — λ_{res}, FSR, IL:** keep in linear (physical-unit) scale; their dynamic range within the defined design space does not warrant a log transform.

2.4 Train / Validation / Test Split

- **Split ratio:** 70% train / 15% validation / 15% test, fixed random seed = 42.
- Split at the level of whole geometry samples, never at the level of individual output targets — this mirrors the design-level splitting discipline in Paper 9, ensuring no information about a given geometry leaks between splits.
- **The test set is identical across all four methods.** This single held-out 15% is never touched during any hyperparameter search (Phases 3–5) and is used exactly once, in Phase 6, to produce the final comparison table.
- For Random Forest and XGBoost hyperparameter tuning specifically, use 5-fold cross-validation within the 70% training pool (matching Paper 6's 4-fold precedent, extended to 5-fold for a slightly larger dataset); for the DNN and Tandem Network, use the 15% validation split directly with early stopping.

2.5 Optional Engineered Features (ablation, not required for the core comparison)

As a secondary experiment only, construct one additional feature set: raw geometry $[R, w, g, L_c]$ augmented with an analytically-estimated coupling coefficient $\kappa(g, L_c)$ from the Phase 1.6 coupling-boundary study. Train each method on both the raw-geometry-only feature set and the augmented feature set, and report whether the engineered feature helps.

[v1.3] **Fix 10 — corrected rationale for this ablation.** Earlier versions described this ablation as "directly inspired by Paper 6's finding that a compact, physically-informed input representation outperformed a naive raw-spectrum input." **Paper 6 reports no such finding.** It uses the integrated dispersion D_{int} (and, for experimental data, its quadratic fit) as the model input throughout, and never trains on or compares against a raw spectrum. The ablation itself is retained unchanged and remains well-motivated — but as *this thesis's own hypothesis*, supported by the general principle that physically-informed features help in small-data regimes, not as a replication of a published Paper 6 result. Do not cite Paper 6 for this claim in the thesis text.

Expected output of Phase 2: `train.csv`, `val.csv`, `test.csv` (identical across all four methods), a fitted normalization/scaling object, a data-quality log, and a coverage-check figure set.

4.3 Phase 3 — Forward Model (Geometry → Response)

Purpose: Build a fast surrogate for the geometry-to-optical-response mapping. This model is (a) a standalone deliverable for accelerating design-space exploration, and (b) the frozen forward half of the Tandem Network in Phase 4.

3.1 Architecture Choice and Justification

A fully-connected feed-forward multilayer perceptron (MLP), not a convolutional or image-based architecture. *Justification:* the input is four low-dimensional continuous scalar geometric parameters, not spatial/pixel data; Paper 2 states that MLPs *are a valid option* for inverse design "when a model is parameterized with only a low number of parameters and/or when the dataset comprises relatively high-level features of low dimensionality", and recommends CNNs for more general parameterizations at the pixel/voxel level — i.e. for free-form, topology-optimization-style design representations, which this project does not use. [v1.3] Fix 15: earlier versions rendered this as Paper 2 stating MLPs are "the appropriate choice"; Paper 2's actual wording is weaker ("can be a valid option"), and it separately warns that fully-connected networks are "particularly prone to overfitting" at large layer widths — a caveat consistent with, and an additional argument for, the deliberately modest architecture fixed below.

3.2 Fixed Default Architecture

- **Input layer:** 4 nodes (R, w, g, L_c, normalized).
- **Hidden layers:** 3 fully-connected layers, widths 64–128–64, Leaky ReLU activation ($\alpha = 0.2$, matching Paper 9's justification: avoids the dying-ReLU problem for continuous regression).
- **Output layer:** 4 nodes (λ_{res}, FSR, $\log_{10} Q_L$, IL, all normalized).
- **Rationale for depth:** a deliberately modest, overfitting-resistant choice appropriate for a dataset of ∼700 training samples, with the final depth selected empirically by the Phase 3.4 search over depth $\in \{2, 3, 4\}$.

[v1.3] **Fix 3 — corrected attribution for the depth rationale.** Earlier versions justified 3 hidden layers by stating that "Paper 3's forward-network depth study found accuracy plateaus beyond ∼4 hidden layers for this class of problem." **Paper 3 reports the opposite trend within its tested range.** Its Supporting Information §II compares Architectures 1–4 with 1, 2, 3 and 4 hidden layers and states that "as the network becomes deeper, the error decreases" — monotonically, up to 4, which is the deepest configuration tested. No plateau beyond 4 is demonstrated, because nothing beyond 4 was tried. The "ultimate performance differs very little" observation in Paper 3 concerns network *width* (Fig. S2b), not depth.

The 3-hidden-layer default is retained unchanged, on the grounds that were always the substantive ones: a ∼700-sample training set, overfitting resistance, and the fact that Phase 3.4 searches depth $\in \{2, 3, 4\}$ and selects empirically. Only the citation has been removed. If the Phase 3.4 search selects depth 4, that is consistent with Paper 3's actual reported trend and should be reported as such rather than treated as anomalous.

3.3 Training Configuration (fixed defaults)

- **Loss:** Mean Squared Error (MSE), computed per output and averaged (equal weighting across the four targets as the default; if one target's error dominates, revisit with weighted MSE and log the change).
- **Optimizer:** Adam, initial learning rate 1×10^{-3}, exponential decay.
- **Batch size:** 16 (small, appropriate for a ∼700-sample training set).
- **Regularization:** L2 weight decay, $\lambda \in [10^{-6}, 10^{-5}]$ (tune within this range); early stopping

on validation loss, patience = 40 epochs.

- **Weight initialization:** Glorot/Xavier uniform. [v1.3] Fix 9: this is the documented practice in **Paper 7**, not Paper 9 as stated in v1.0–v1.2. Chen & Jiang initialize all weights from a uniform distribution on $[-\text{limit}, +\text{limit}]$ with limit $= \sqrt{6/(\text{fan_in} + \text{fan_out})}$ and all biases to zero — that is precisely Glorot uniform. Paper 9 does not document its weight initialization anywhere in its main text. The default is unchanged; the citation has been moved to the paper that actually supports it.

3.4 Hyperparameter Search

Grid or random search over the following four axes:

- hidden-layer depth $\in \{2, 3, 4\}$;
- hidden-layer width $\in \{32, 64, 128, 256\}$;
- learning rate $\in \{10^{-4},\ 5 \times 10^{-4},\ 10^{-3},\ 5 \times 10^{-3}\}$;
- L2 weight decay $\in \{10^{-6},\ 10^{-5},\ 10^{-4}\}$.

Select the configuration minimizing validation MSE. Report the full search grid and the selected configuration in the thesis appendix (non-negotiable for reproducibility).

3.5 Step-by-Step Procedure

1. Load `train.csv` / `val.csv` (Phase 2 output).
2. Instantiate the MLP with the Phase 3.2 default architecture.
3. Run the Phase 3.4 hyperparameter search, tracking validation MSE for each configuration.
4. Retrain the best configuration on the full 70% training pool with early stopping against the validation set.
5. Save the trained forward model's weights — this exact, frozen model is reused unmodified as f_θ in Phase 4.
6. Evaluate on the test set (Phase 6 metrics) and report per-target R^2, MAPE, RMSE.

Expected output of Phase 3: A trained, frozen forward MLP; a hyperparameter-search log; per-target test-set accuracy (R^2, MAPE, RMSE); training/validation loss curves.

4.4 Phase 4 — Inverse Model and Tandem Network Design

Purpose: Build two inverse-direction models — a naive direct-inverse DNN (baseline) and a Tandem Network (the field's established fix for the one-to-many/non-uniqueness problem) — and use their comparison to directly answer Objective O6.

4.1 Why Naive Inverse Networks Can Fail — the Underlying Math

Let $F : D \mapsto R$ be the true geometry-to-response mapping. F is **deterministic and single-valued** — every geometry has exactly one response — but it is **not injective**: distinct geometries $D_A \neq D_B$ can produce nearly identical responses, $F(D_A) \approx F(D_B) = R_0$. In the standard terminology of this literature, F is a **many-to-one** mapping. This is a real possibility in a four-parameter MRR space where, e.g., a larger gap combined with a longer coupling length can reproduce the coupling strength of a smaller gap with a shorter coupling length.

[v1.3] **Fix 1 — corrected characterisation of F.** Versions 1.0–1.2 described F as "(deterministic, one-to-one)" in the same sentence that asserted $F(D_A) \approx F(D_B)$ for $D_A \neq D_B$ — a direct self-contradiction, since a mapping that sends two distinct inputs to the same output is by definition not one-to-one. The correct statement is *deterministic and single-valued but not injective*, i.e. many-to-one; this matches the terminology of both primary sources (Liu et al. [3] Supporting Information §I: "a nonlinear operator $\hat{O}$ defines a many-to-one mapping"; Adibnia et al. [9], which uses "many-to-one" throughout). The mathematics of Eqs. 8–9 and the argument they support are unaffected — the many-to-one property is what *creates* the problem the tandem network solves, so the corrected wording strengthens rather than weakens the argument. This error sat in the opening sentence of the derivation underpinning Objective O6, the thesis's primary contribution, and must be corrected before the Related Work and Methodology chapters are drafted from it.

A naive inverse network $g_\phi : R \mapsto D$ trained with the direct parameter-matching loss

$$\mathcal{L}_{naive}(\phi) = \frac{1}{N}\sum_{i=1}^{N} \left\| g_\phi(R_i) - D_i \right\|^2 \tag{8}$$

receives, for inputs near R_0, conflicting gradient signals pulling toward D_A in one training instance and toward D_B in another. Paper 3 shows this data-inconsistency problem persists even after removing explicit duplicate-response instances, because implicit near-duplicates remain (their filtering experiment retained 26.1% of instances and barely improved training); training can stall or converge to an unphysical average of D_A and D_B.

4.2 The Tandem Network Fix — the Underlying Math

Let f_θ be the Phase 3 forward network, with weights θ frozen after training. Define the tandem network as the composition $f_\theta \circ g_\phi$: the inverse network g_ϕ proposes a geometry $\hat{D} = g_\phi(R)$, which is then passed through the frozen forward network to reconstruct a response $\hat{R} = f_\theta(\hat{D})$. Only ϕ is trained; θ is not updated. The tandem loss is computed in response space, not parameter space:

$$\mathcal{L}_{tandem}(\phi) = \frac{1}{N}\sum_{i=1}^{N} \left\| f_\theta\big(g_\phi(R_i)\big) - R_i \right\|^2 \tag{9}$$

This is the precise mechanism that resolves non-uniqueness: if $g_\phi(R_0)$ converges to D_A or D_B *or any other geometry whose forward-predicted response matches* R_0, the loss is equally low in every case. The network is never penalized for finding a valid solution that differs from whichever specific geometry happened to generate that response in the training set — eliminating the conflicting-gradient pathology in Eq. 8. The documented trade-off is that the tandem network will converge to only one of the possibly many valid solutions, not enumerate all of them; this is an accepted limitation, not a defect to engineer around within this thesis's scope.

[v1.3] **Fix 16 — attribution precision on the single-solution trade-off.** Earlier versions attributed this trade-off to "Papers 1, 2, 3" and, in Section 12.1, stated that Wiecha et al. [1] "are explicit" about it. Paper 1 makes the point by *contrast* rather than by direct statement: it presents conditional GANs, conditional variational autoencoders, and mixture-density networks as the architectures that can "address the distinct non-unique solutions" and "deliver all possible solutions together with a weight for their respective priorities", while the tandem network is described only as making "all correct solutions yield a positive training feedback". The trade-off is a correct and uncontroversial reading of Paper 1, and remains the accepted position in the field — but it should be presented as an inference from Paper 1's taxonomy rather than as a quoted claim.

4.3 Architecture (both models)

Same MLP topology as the forward model (Phase 3.2) with input/output reversed: input layer 4 nodes (normalized λ_{res}, FSR, $\log_{10} Q_L$, IL), 3 hidden layers 64–128–64 (Leaky ReLU, $\alpha = 0.2$), output layer 4 nodes (normalized R, w, g, L_c). Using the same base topology for both the naive inverse network and the tandem network's inverse half isolates the comparison to the loss function/training procedure, not architecture size — required for a clean test of Objective O6.

4.4 Step-by-Step Procedure

1. **Naive inverse baseline:** train g_ϕ directly against Eq. 8 using the same optimizer, regularization, and early-stopping settings as Phase 3.3. Evaluate against ground-truth test geometries (R^2, MAPE per parameter, RMS-RGE). [v1.3] This is the *geometry-space* half of Objective O6's dual criterion.
2. **Tandem network:** load the frozen Phase 3 forward model as f_θ; initialize a fresh inverse network g_ϕ (same architecture as 4.3, different random initialization than the naive baseline); train using Eq. 9, backpropagating through the frozen f_θ but updating only ϕ. Evaluate the reconstructed response $f_\theta(g_\phi(R_i))$ against the true target R_i, and separately evaluate the predicted geometry $g_\phi(R_i)$ against the ground-truth geometry D_i purely for diagnostic comparison against the naive baseline (expect this second number to be less favorable than the naive baseline's own optimized parameter-matching accuracy, by construction — report and discuss this honestly, do not treat it as a failure).
3. **Optional physics-informed loss extension** (stretch goal, run only after items 1–2 are complete and documented): add a regularization term using the analytical resonance relation (Eq. 1/6) evaluated at the predicted geometry $\hat{D}_i$:

$$\mathcal{L}_{phys}(\phi) = \frac{1}{N} \sum_{i=1}^{N} \left(\lambda_{res}^{analytic}(\hat{D}_i) - \lambda_{res,i}^{target} \right)^2 \tag{10}$$

$$\mathcal{L}_{total}(\phi) = \mathcal{L}_{tandem}(\phi) + \beta \, \mathcal{L}_{phys}(\phi), \qquad \beta \in \{0.1, 0.5, 1.0\} \tag{11}$$

   Run the ablation with $\beta = 0$ (no physics term) vs. $\beta > 0$, matching Paper 12's ablation design, and report whether the physics-informed term improves test-set generalization even if it does not improve training loss.
4. **Closed-loop re-verification (mandatory, not optional):** for a representative subset of test-set targets ($n \geq 20$), take each model's predicted geometry, re-run it through Lumerical FDTD (or the MEEP fallback), and compare the re-simulated response to the originally requested target — following the predict → re-simulate → compare discipline established in Papers 7, 9, and 12. This is the ultimate validation and must not be skipped even if time is tight; if compute time is the binding constraint, reduce n in this step before reducing scope elsewhere. [v1.3] This is the *response-space* half of Objective O6's dual criterion, and it is the criterion under which Papers 7 and 9 reported their naive inverse networks as converging acceptably. Report the naive baseline's closed-loop error alongside its geometry-space R^2 whenever O6 is discussed.

Expected output of Phase 4: Trained naive-inverse DNN; trained tandem network (and, optionally, its physics-informed variant); a direct, quantified answer to whether the one-to-many problem manifests in this MRR design space, reported against both the geometry-space and response-space criteria (Objective O6)[v1.3]; closed-loop re-verification results for ≥ 20 representative targets.

4.5 Phase 5 — Baseline Comparison Models (Random Forest / XGBoost)

Purpose: Fill the gap identified in Papers 6, 7, 9, 10, and 12: no prior microring-resonator inverse-design study benchmarks classical ensemble ML against DNN/tandem architectures.

5.1 Models and Multi-Output Strategy

- **Random Forest Regressor** (scikit-learn): use native multi-output support — a single `RandomForestRegressor` predicting all four targets at once; report per-target metrics regardless.
- **XGBoost Regressor** (`xgboost` package): wrap `XGBRegressor` in `sklearn.multioutput.MultiOutputRegressor`, since native multi-output support is more limited than Random Forest's.
- Train one Forward RF, one Forward XGBoost, one Inverse RF, one Inverse XGBoost — four models total, evaluated on the identical Phase 2 test split.

5.2 Hyperparameter Grids

Random Forest Grid		XGBoost Grid	
`n_estimators`	$\{100, 200, 300, 500\}$	`n_estimators`	$\{100, 300, 500\}$
`max_depth`	$\{10, 20, 40, 80, \text{None}\}$	`max_depth`	$\{3, 6, 9\}$
`min_samples_leaf`	$\{1, 2, 5, 7\}$	`learning_rate`	$\{0.01, 0.05, 0.1, 0.3\}$
`max_features`	$\{\text{sqrt}, \text{log2}, \text{None}\}$	`subsample`	$\{0.7, 0.85, 1.0\}$
		`colsample_bytree`	$\{0.7, 0.85, 1.0\}$
		`reg_lambda` (L2)	$\{0, 1, 5\}$

Use `GridSearchCV` (or `RandomizedSearchCV` with ≥ 60 iterations if the full grid is too large for the available compute budget — log which was used) with 5-fold cross-validation on the 70% training pool, minimizing mean validation MAPE across all four targets.

5.3 Normalization Test (mandatory, not optional)

Run every RF and XGBoost model on both normalized and unnormalized inputs/targets, and report both results side by side. Paper 6's Table 1 shows normalization degraded Random Forest accuracy on one target (radius, 12.52 → 21.43 MAPE) while improving it on the other two (width, 3.14 → 1.89; height, 15.01 → 6.34) — let the data decide, and report the empirical finding either way. [v1.3] See the Phase 2.3 note on the discrepancy between Paper 6's Table 1 and its own prose summary.

5.4 Step-by-Step Procedure

1. Load `train.csv` (both normalized and raw versions) and `val.csv`.
2. For each of the four models and each preprocessing condition, run the Phase 5.2 hyperparameter search via 5-fold CV.
3. Select the best configuration per model per preprocessing condition (lowest mean CV MAPE).
4. Retrain the selected configuration on the full 70% training pool.
5. Evaluate on the identical Phase 2 test set (Phase 6 metrics).
6. Record feature importances (Random Forest's native importance, XGBoost's gain-based importance) for the forward-direction models.

Expected output of Phase 5: Four tuned models, each with normalized and raw-input variants; a hyperparameter table; feature-importance rankings; per-target test-set metrics ready for Phase 6.

4.6 Phase 6 — Evaluation Metrics

Purpose: Apply one fixed, identical set of metrics to all four methods on the identical held-out test set, avoiding the apples-to-oranges cross-paper benchmarking problem visible in Paper 12's own comparison table (which places a spectral-accuracy percentage, a Chamfer distance, a dispersion improvement over PSO, an RMSE from a 4-D flow-MRI study, and an RMS-RGE side by side as though they were commensurable).

6.1 Per-Target Regression Metrics

For each of the four targets (λ_{res}, FSR, Q_L, IL), report all three of the following (never only one):

$$\text{MAPE} = \frac{100\%}{n}\sum_{i=1}^{n}\left|\frac{x_i^{pred} - x_i^{act}}{x_i^{act}}\right| \tag{12}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(x_i^{pred} - x_i^{act}\right)^2} \tag{13}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(x_i^{act} - x_i^{pred}\right)^2}{\sum_{i=1}^{n}\left(x_i^{act} - \bar{x}^{act}\right)^2} \tag{14}$$

6.2 Inverse-Design-Specific Metrics

RMS Relative Geometric Error — a single aggregate summary across all four geometric parameters, expressed as a percentage:

$$\text{RMS-RGE} = 100\% \times \sqrt{\frac{1}{4}\sum_{p\in\{R,w,g,L_c\}}\left(\frac{p^{pred} - p^{target}}{s_p}\right)^2} \tag{15}$$

[v1.3] **Fix 4 — two corrections to RMS-RGE, both required for Objective O3 to be well defined.**

(a) The 100%× factor. As typeset in v1.0–v1.2 the expression returned a *fraction*, while Section 10.5 lists its units as % and Objective O3 sets a $\leq 15\%$ threshold. Paper 12, the source of this metric, reports it as a percentage (3.46% FDTD, 5.14% analytical). The factor is now explicit. *No threshold has changed* — O3's $\leq 15\%$ is unchanged and now unambiguous.

(b) The normalizer s_p, and why p^{target} alone will not do. Paper 12 normalises each term by its target value, which is safe in *its* design space because every parameter is strictly positive and bounded away from zero. **This thesis's design space includes $L_c = 0$ as an explicitly valid design point** (Phase 1.2), where a target-normalised relative error is undefined, and where small L_c targets produce arbitrarily large relative errors that would dominate the aggregate. Fixed convention:

- For R, w, g (strictly positive, bounded away from zero): $s_p = p^{target}$, i.e. the conventional relative error, identical to Paper 12.
- For L_c: $s_p =$ the *range width* of L_c, i.e. $3\,\mu$m. This is a range-normalised absolute error, well defined at $L_c = 0$.

Reporting requirement: report the aggregate RMS-RGE under this convention as the headline number for O3, and *additionally* report the four per-parameter error terms separately so that the L_c convention is fully transparent and a reader can recompute under any alternative. Also report the pure Paper-12-convention value computed over $\{R, w, g\}$ only, for direct cross-paper comparability with Paper 12's 3.46%/5.14%.

Closed-loop response error — for each of $n \geq 20$ representative test-set targets, re-simulate the predicted geometry and compute the percentage deviation between the re-simulated response and the originally requested target response, per target quantity.

6.3 Sample-Efficiency Curve

Train all four methods on progressively larger random subsets of the 70% training pool (sizes 100, 200, 400, 600, and the full pool — minimum 5 points), evaluate each on the fixed test set, and plot test MAPE (aggregate) vs. training-set size for all four methods on one figure.

[v1.2] **Fix 1 — mandatory verbal verdict (closes the Objective O5 traceability gap):** Producing the figure is not sufficient on its own. After plotting the curve, state in writing, as an explicit sentence in the results text (not left for the reader to read off the figure): **(a) which method achieves the lowest aggregate test MAPE in the ≤ 300-sample regime, and (b) which method achieves the lowest aggregate test MAPE at the full dataset size**, matching Objective O5's exact wording. This mirrors the mandatory-verdict discipline already required by Phase 6.4 ("every method appears in every metric table") and Phase 7.4 ("state explicitly whether… meets the scenario's tolerance"): an artifact (the curve) is not itself the deliverable Objective O5 asks for — the written verdict is.

[v1.2] **Fix 4b — framing:** this curve is this thesis's direct, honest answer to the question a Q1/Q2 reviewer would immediately raise about a few-hundred-to-1,500-sample dataset: not "is this dataset large enough in absolute terms," but "how does each method's accuracy actually degrade as the dataset shrinks, measured rather than assumed." Present it as that explicit answer in the thesis text, not only as a supporting figure. [v1.3] Two concrete comparison anchors worth stating alongside it: Paper 6 obtained its headline result from **462** FEM samples, and Paper 12 — the most recent work reviewed — from **2,500** geometries with an 80/20 split and 80 training epochs. A measured sample-efficiency curve over 100–1,500 samples is therefore squarely within, not below, the operating range of the closest published work.

6.4 Reporting Discipline (mandatory)

- Every method appears in every metric table, even where it performs worst.
- Report per-target results and one aggregate summary (mean MAPE across targets; RMS-RGE).
- State explicitly, for at least one parameter, whether it is harder to predict than the others (the coupling gap g is hypothesized, by analogy with Paper 6's finding that radius was its hardest parameter, to be the hardest here) — confirm or refute with actual results, do not assume it.

Expected output of Phase 6: One master results table; the sample-efficiency figure with its accompanying written ≤300-sample/full-dataset verdict[v1.2]; the closed-loop validation report; an explicit per-parameter difficulty ranking with discussion; and the per-parameter RMS-RGE breakdown required by Phase 6.2[v1.3].

4.7 Phase 7 — Application-Specific Validation (CPO / Interconnect Mapping)

Purpose: Translate raw regression accuracy into concrete co-packaged-optics design implications, directly delivering the thesis's stated differentiator versus generic photonics ML benchmarks (Objective O7).

7.1 Bandwidth / Channel-Count Relations

$$\Delta f_{3dB} = \frac{c\,\Delta\lambda_{FWHM}}{\lambda_{res}^2} \tag{16}$$

$$N_{ch} \approx \frac{\mathrm{FSR}}{\Delta\lambda_{ch}} \tag{17}$$

7.2 Cascaded Insertion-Loss Budget

$$\mathrm{IL}_{total}\,[\mathrm{dB}] = \sum_{i=1}^{N} \mathrm{IL}_i \approx N \cdot \mathrm{IL}_{avg} \tag{18}$$

[v1.3] **Fix 6 — what Eq. 18 is and is not (clarification; the Scenario B threshold is unchanged).** Eq. 18 is a **deliberate worst-case bound**, and the thesis text must say so rather than presenting it as the physical loss experienced by a given channel. In a real N-ring demultiplexer, a channel that drops at ring k experiences *one* drop-port insertion loss plus $(k-1)$ off-resonance through-port transits, not N drop-port losses. For $N = 8$, $\mathrm{IL}_{drop} = 1.0\,\mathrm{dB}$ and a typical off-resonance through-port loss of $\sim 0.1\,\mathrm{dB/ring}$, the worst channel accumulates roughly $1.0 + 7 \times 0.1 \approx 1.7\,\mathrm{dB}$ — not 8 dB. Eq. 18 therefore overstates the per-channel loss by roughly a factor of four.

The 1.0 dB/ring Scenario B threshold is retained unchanged, for two reasons: (i) it is a conservative bound and conservatism is appropriate for a link budget at design time; (ii) relaxing it would weaken a fixed default on the basis of an assumed through-port loss figure that this thesis does not measure. What changes is only the *stated justification*: Scenario B's IL requirement is a worst-case cascade bound, not a per-channel prediction. If the Phase 1 dataset yields reliable off-resonance through-port transmission values, report the refined per-channel budget alongside the Eq. 18 bound as a secondary result.

7.3 Two Worked CPO Scenarios (fixed, non-ambiguous targets)

Scenario A — Moderate-density CPO link (4 channels, wide grid). Target $\lambda_{res} = 1550\,\text{nm}$; channel spacing 4 nm; required FSR $\geq 16\,\text{nm}$; target FWHM 0.2–1.0 nm (i.e. $Q_L \approx 1{,}500$–$8{,}000$); IL $\leq 1.5\,\text{dB}$ per ring.

[v1.3] Fix 7b — **achievability check (numbers now stated, target unchanged).** At the nominal $n_g = 4.2$ fixed in Phase 1.2, FSR $\geq 16\,\text{nm}$ requires $L \leq 35.75\,\mu\text{m}$, i.e. $R \lesssim 5.69\,\mu\text{m}$ at $L_c = 0$, tightening as L_c grows. Earlier versions stated this as "achievable near $R \approx 5$–$6\,\mu$m"; the accurate statement is $R \approx 5.0$–$5.7\,\mu\text{m}$ at $L_c = 0$. Two consequences to note in the thesis text: Scenario A occupies only the lowest ∼7% of the R range, so under LHS with $N = 1{,}000$ roughly 70 training samples will fall in it, and inverse queries for Scenario A therefore sit close to the boundary of the training distribution. Report this explicitly when discussing Scenario A results — it is a legitimate and interesting finding if boundary proximity degrades accuracy, and it is exactly the kind of effect Paper 9 probed deliberately with its "furthest data" test.

Scenario B — Dense CPO link (8 channels, 100 GHz/0.8 nm ITU-like grid). Target $\lambda_{res} = 1550\,\text{nm}$; channel spacing 0.8 nm; required FSR $\geq 6.4\,\text{nm}$; target FWHM $\leq 0.2\,\text{nm}$ (i.e. $Q_L \gtrsim 7{,}750$); IL $\leq 1.0\,\text{dB}$ per ring, since Eq. 18 implies a 1.0 dB/ring budget keeps an 8-ring demux chain within an 8 dB total link-budget allowance *under the worst-case cascade bound described above*[v1.3].

[v1.3] Achievability check: FSR $\geq 6.4\,\text{nm}$ requires $L \leq 89.4\,\mu\text{m}$, i.e. $R \lesssim 14.2\,\mu\text{m}$ at $L_c = 0$ — comfortably inside the Phase 1.2 range. $Q_L \gtrsim 7{,}750$ follows directly from 1550/0.2 via Eq. 5 and is confirmed. Both scenarios therefore require Q_L only in the 1.5×10^3–8×10^3 band, which is unaffected by the Section 2.3 scope note on the upper end of the moderate-Q range.

7.4 Step-by-Step Validation Procedure

1. For each scenario, construct the target tuple (λ_{res}, FSR, Q_L, IL).
2. Feed each target into every trained inverse model to obtain four candidate geometries.
3. Re-simulate all four candidate geometries (closed-loop procedure, extended to these CPO-labelled targets).
4. For each method, state explicitly whether the re-simulated response meets the scenario's stated tolerance on every spec, or identify which spec(s) it misses and by how much.
5. Translate any IL or FWHM miss into its concrete CPO consequence.

7.5 Thermal Tunability (qualitative extension, explicitly out of core scope)

$$\frac{d\lambda_{res}}{dT} \approx \frac{\lambda_{res}}{n_g} \frac{dn_{eff}}{dT} \tag{19}$$

using the silicon thermo-optic coefficient ($dn/dT \approx 1.86 \times 10^{-4}\,\text{K}^{-1}$, giving a typical $d\lambda_{res}/dT$ on the order of 70–90 pm/K for silicon strip waveguides at 1550 nm). If time allows, report the tuning range needed to compensate a representative fabrication-induced resonance offset.

[v1.3] Note: Eq. 19 requires dn_{eff}/dT (the *modal* thermo-optic coefficient), whereas the figure quoted is the *material* dn/dT for silicon. The two coincide only at unity confinement. Sanity check at $n_g = 4.2$: $(1550/4.2) \times 1.86 \times 10^{-4} \approx 69\,\text{pm/K}$, so the quoted 70–90 pm/K band is correct at its lower end and slightly optimistic at its upper end. Use the mode-solver's own dn_{eff}/dT if this extension is attempted.

Expected output of Phase 7: A scenario-based validation table; at least one fully worked numerical link-budget translation; an optional thermal-tunability discussion.

5 Expected Results / Success Criteria

Phase	Good result looks like	Revision trigger
1. Dataset	$N_{clean} \geq 800$; literature-validation FSR error $\leq 10\%$; no obvious coverage gap.	Validation error $> 15\%$ (re-check mesh/material model); $> 10\%$ of samples flagged/excluded.
2. Preprocessing	Clean, balanced marginal distributions.	Gap in the high-Q_L/low-IL tail persists after one supplementary batch — narrow the affected range.
3. Forward model	$R^2 \geq 0.90$, MAPE $\leq 10\%$ on ≥ 3 of 4 targets.	Validation loss plateaus far above training loss — increase regularization before assuming insufficient data.
4. Inverse/-Tandem	Tandem's reconstructed-response MAPE is competitive with the forward model's own accuracy.	Tandem reconstructed-response error worse than naive DNN's forward consistency — re-check frozen weights.
5. RF/XGBoost	Comparable or superior MAPE to the DNN in the ≤ 300-sample regime.	Grid search selects boundary hyperparameters every time — widen the grid.
6. Evaluation	One consistent table; clear sample-efficiency ranking; RMS-RGE $\leq 15\%$ for ≥ 1 method.	No method achieves RMS-RGE $\leq 25\%$ — revisit design-space redundancy.
7. CPO validation	≥ 1 method meets all four specs in ≥ 1 scenario.	No method meets IL or Q tolerance — revisit Phase 1.6 coupling-boundary check.

6 Tools & Software Stack

Tool	Used for
Ansys Lumerical FDTD (3-D)	Phase 1 dataset generation (primary solver); Phase 4.4/6.2/7.4 closed-loop re-verification. [v1.1] See Section 7 for the MEEP/Tidy3D contingency if no license is available.
Lumerical Python API (`lumapi`)	Scripted, unattended batch simulation across all ≥ 800 samples; not the GUI.
MEEP (open-source FDTD)[v1.1]	Contingency solver if no Lumerical license exists; validated for this device class by Chen & Jiang [7].
Tidy3D (Flexcompute)[v1.1]	Secondary contingency; free-tier academic "Education Free" plan.
Python 3.x	All orchestration, data processing, and modelling code.
NumPy, Pandas	Numerical array handling and the tabular dataset pipeline.
scikit-learn	Random Forest; `MultiOutputRegressor`; `GridSearchCV`/ `RandomizedSearchCV`; k-fold CV; scaling; metrics.
XGBoost (Python package)	Gradient-boosted tree regressor for the classical-ML baseline.
PyTorch	Forward MLP, naive inverse DNN, and Tandem Network — chosen for its explicit, manual training-loop control needed for the frozen-forward-model tandem step.
Matplotlib / Seaborn	All figures: EDA distributions, training curves, sample-efficiency curves, scenario validation plots.
Git / GitHub	Version control and code/dataset release, following the open-release precedent set by Papers 9 and 12. [v1.2] See **Phase 1.5 step 8** for the mandated publish action this table entry corresponds to.[v1.3]
Jupyter notebooks	Interactive development; convert to scripts for the final, reproducible batch pipeline.
LaTeX (Overleaf or local)	Thesis writing and this methodology document.

[v1.3] **Fix 5 — cross-reference correction.** This row, and the v1.2 Version History entry that introduced it, both referred to "Phase 1.5 step 9". Phase 1.5 contains **eight** numbered steps; the v1.2 dataset-publication action is **step 8**. Both references have been corrected. No content change — the mandated action itself is unchanged.

7 Risk / Limitation Log

Risk	Impact	Mitigation / Fallback
FDTD runtime too slow to reach $N = 1{,}000$ within the timeline.	High	Fall back to the $N = 800$ minimum (Objective O1); if still short, generate a supplementary cheap analytical/CMT dataset alongside the FDTD set (dual-fidelity strategy), clearly labelling each sample's origin. [v1.1] If the Phase 1.6 timing-calibration gate shows > 25 min/sample even at reduced mesh settings: downgrade to 2-D FDTD for the bulk of the dataset (precedent: Chen & Jiang [7]), retaining a smaller ($n \approx 100$–150) 3-D-simulated subset spanning the full parameter range specifically to quantify the systematic 2-D-vs-3-D offset on each target quantity. Report this offset explicitly and either (a) apply it as a correction to the 2-D bulk dataset, or (b) report 2-D-only results with the 3-D-derived offset stated as an explicit accuracy bound — do not present 2-D results as equivalent to 3-D without this check.
Lumerical license/seat-time availability limited.	Medium	Queue simulations overnight/off-hours; batch via `lumapi` scripts, never manual GUI runs; if severely constrained, reduce mesh resolution slightly after re-running the Phase 1.6 convergence check at the reduced resolution.
[v1.1] No institutional Lumerical license available at all (distinct from limited seat-time).	High	Switch Phase 1's solver to MEEP (open-source FDTD), already validated for this exact device class — a bus-coupled, multi-ring channel-dropping filter — by Chen & Jiang [7], who generated 25,000 samples in 2-D Meep FDTD for a structurally near-identical problem. Secondary option: Tidy3D (Flexcompute), free-tier academic "Education Free" plan (10 FlexCredits/month at no cost, 1 FlexCredit $\approx$ 50 CPU-core-hours); apply for free trial credits first if not eligible for the academic tier. Either fallback keeps every other phase of this Methodology Plan unchanged (see tool-substitution notes, Section 4.1).
Dataset too small for stable DNN/Tandem training (high run-to-run variance).	Medium	Aggressive early stopping and L2 regularization (Phase 3.3); train each final model with the 5-seed protocol (Section 14) and report mean $\pm$ standard deviation, not a single run.
One-to-many ambiguity turns out more severe than expected; tandem network still struggles.	Medium	This is itself a valid, reportable scientific finding (Objective O6 is designed to accept either outcome); if it blocks Phase 6/7 progress, narrow the g/L_c ranges to reduce redundancy and document the narrowing explicitly.
Coupling gap g is highly non-linear and the hardest parameter to predict.	Low–Medium	Acceptable if it happens — report and discuss per Phase 6.4; if MAPE for g exceeds 50%, try a log or reciprocal transform on g specifically.
No fabrication access $\Rightarrow$ unavoidable sim-to-real gap	Medium (accepted, not	Explicitly scope the thesis as simulation-only; substitute the Phase 1.6 literature-validation step as the closest practical stand-in. [v1.3] This is the single largest constraint on the venue tier this work can

8 Timeline

A 24-week default schedule, [v1.1] explicitly assuming a full-time-equivalent pace of 35 hours/week ($24 \times 35 = 840$ total hours) — comparable to a full-time research placement, not a part-time BSc schedule alongside coursework. See the part-time conversion below if that assumption does not match your situation.

Full-Time Schedule (840 total hours assumed)

Weeks	Task	Phase	Deliverable
1–2	Finalize parameter-range physics justification; build `lumapi` template; run mesh-convergence & validation checks.	1 (setup)	Validated template + reports.
3–6	Full LHS-sampled dataset generation run ($N = 800$–$1{,}500$), unattended/batched.	1 (exec.)	Raw dataset + quality log.
7	Cleaning, EDA, coverage check, normalization, train/val/test split.	2	`train/val/test.csv`.
8–9	Forward MLP development and hyperparameter search.	3	Trained, frozen forward model.
10–12	Naive inverse DNN; Tandem network training; closed-loop re-verification.	4	Trained inverse models + validation results.
13–14	Random Forest and XGBoost training and hyperparameter search.	5	Tuned RF/XGBoost models.
15	Unified metric computation; sample-efficiency curves.	6	Master results table + figure.
16–17	CPO scenario construction, inverse queries, closed-loop re-verification.	7	Scenario validation table.
18–20	Results write-up, discussion, limitations.	—	Draft chapters.
21–22	Full draft assembly; supervisor feedback loop.	—	Complete draft.
23–24	Revisions; final formatting; submission.	—	Final thesis.

[v1.1] **Fix 4 — Part-Time Equivalent Schedule:** At 15–20 hrs/week (midpoint 17.5, i.e. half the assumed full-time rate): $840 \div 17.5 = 48$ weeks (≈11 months) for the identical total effort, same phase order and dependencies — only the calendar axis changes (every week-range below is doubled from the table above).

Part-Time Equivalent Schedule (15–20 hrs/week, ≈48 weeks total)

Weeks (part-time)	**Task**	**Phase**	**Deliverable**
1–4	Setup: parameter-range justification, template build, convergence & validation checks.	1 (setup)	Validated template + reports.
5–12	Full dataset generation run.	1 (exec.)	Raw dataset + quality log.
13–14	Cleaning, EDA, split.	2	`train`/`val`/`test` CSVs.
15–18	Forward MLP + search.	3	Frozen forward model.
19–24	Inverse DNN, Tandem, closed-loop check.	4	Trained inverse models.
25–28	RF/XGBoost + search.	5	Tuned RF/XGBoost models.
29–30	Unified metrics, sample-efficiency curve.	6	Master results table.
31–34	CPO scenarios, closed-loop, link-budget translation.	7	Scenario validation table.
35–40	Results write-up.	—	Draft chapters.
41–44	Draft assembly, feedback.	—	Complete draft.
45–48	Revisions, submission.	—	Final thesis.

One caveat, not a further change: Weeks 5–12 (Phase 1 execution) are governed primarily by wall-clock compute time (Fix 3, Section 4.1), not by the student's personal weekly hours — once queued, unattended batch runs proceed regardless of part-time status. The doubled duration shown is a reasonable scheduling default (setup, monitoring, and re-queuing still need part-time hours), but if compute genuinely runs unattended and uninterrupted, this specific span could compress; use the Phase 1.6 timing-calibration result to adjust this row if it materially differs from the 8 hrs/day assumption used there.

9 Reference Table — All 12 Papers Mapped to This Methodology

Papers are numbered exactly as in the original 12-paper literature review. Gap-defining papers (6, 7, 9, 10, 12) are marked with a dagger (†).

#	Authors (Year)	Role	Sec./Phase	What it provides to this thesis
1	Wiecha et al. (2020/21)	Methodology	Sec. 2; Ph. 4	Conceptual foundation for the tandem-network mechanism and the one-to-many problem; general review scope confirms absence of classical-ML comparison field-wide. [v1.3] Its taxonomy of cGAN/cVAE/mixture-density architectures is the basis for the tandem single-solution trade-off (Phase 4.2, Fix 16).
2	Khaireh-Walieh et al. (2023)	Methodology + Gap	Sec. 2; Ph. 1,3,4	Explicit statement that classical ML (RF/SVM) is out of scope for the field's own tutorial — direct textual gap evidence; dataset-size guidance; MLP-vs-CNN justification (see Phase 3.1 wording correction[v1.3]); tandem training procedure.
3	Liu et al. (2018)	Methodology	Ph. 4; Sec. 7	Originating tandem-network paper; source of Eqs. 8–9; motivates Objective O6. [v1.3] Its SI §II forward-depth study is *not* a plateau result — see Phase 3.2, Fix 3.
4	Tahersima et al. (2018/19)	Methodology	Ph. 1,6; Sec. 7,8	Lumerical-FDTD dataset-scale precedent (∼20,000 samples) calibrating Phase 1's scale-down; correlation-metric precedent (a Pearson correlation coefficient $> 99\%$, not R^2[v1.3]); dataset-coverage-bias warning (their training data skews to high-transmission devices).
5	Yeung et al. (2022)	Methodology	Ph. 1; Sec. 7	Lumerical + Python (`lumapi`/LumOpt) automated-pipeline precedent; independent confirmation of the dataset-coverage-gap risk ("poorly represented by the training dataset… poor starting designs").
6†	Pal et al. (2022/23)	Primary baseline / Gap	Sec. 2; Ph. 2,5,6	Closest prior classical-ML microresonator paper (462 FEM samples, Si_3N_4, no bus waveguide); source of RF/DT hyperparameter-grid template, k-fold CV practice, normalization test, MAPE reporting discipline (its Eq. 4 is this thesis's Eq. 12). Its lack of coupling-gap parameter and DNN/tandem comparison is the central gap this thesis fills. [v1.3] It is *not* a source for the Phase 2.5 engineered-feature rationale (Fix 10).
7†	Chen & Jiang	Primary	Sec. 2; Ph.	Closest prior bus-coupled MRR

Reading Guide

- **Introduction/Related Work chapter:** lead with Papers 6, 7, 9, 10, 12 (†) — these define the precise, triangulated gap. Use Papers 1, 2, 3 for the tandem-network mechanism, Papers 4, 5 for Lumerical-based precedent, Paper 8 for physics-informed bounds and the corrected Q_L formula, Paper 11 to pre-empt scope confusion and to supply the nominal SOI n_g.
- **Methodology chapter:** every phase in Section 4 cites its governing papers inline.
- **Discussion chapter:** the open tension between Paper 3 (naive inverse networks fail) and Papers 7/9 (naive inverse networks work in differently-bounded design spaces) is this thesis's opportunity to contribute a genuinely new, empirically-grounded answer (Objective O6). **[v1.3]** Frame the tension around *response-vector information content* (200/100/800 spectral points vs. this thesis's four scalars) as well as design-space size — see Section 1.2, Fix 14a.

Part II — Supporting Chapters & Reference Appendix

10 Notation and Symbols Table

Every symbol used across Phases 1–7, consolidated into one lookup table, grouped by where it first appears. A disambiguation note at the end resolves symbols deliberately reused with a different meaning in different phases (standard practice where photonics and ML literatures each have their own strong conventions).

10.1 Geometric Design Parameters (Phase 1)

Symbol	Definition	Units
R	Ring radius (free design parameter)	μm
w	Waveguide width (free design parameter)	nm
g	Coupling gap, both couplers (free design parameter)	nm
L_c	Coupling length, straight section of each coupler	μm
h	Waveguide height/thickness (fixed, not swept)	nm

10.2 Optical and Coupled-Mode-Theory Quantities (Phase 1)

Symbol	Definition	Units
λ	Wavelength (general/running variable)	nm
λ_{res}	Resonance wavelength (target/output quantity)	nm
n_{eff}	Effective refractive index of the guided mode	dimensionless
n_g	Group index of the guided mode (nominal value 4.2, Phase 1.2[v1.3])	dimensionless
β	Propagation constant (Phase 1 meaning — see collision note)	rad/μm
ϕ	Round-trip phase accumulated in the ring (Phase 1 meaning)	rad
L	Total round-trip length, $L = 2\pi R + 2L_c$	μm
t_1, t_2	Self-coupling coefficients of the two couplers	dimensionless (0–1)
κ	Cross-coupling coefficient; $\kappa^2 + t^2 = 1$ for a lossless coupler	dimensionless (0–1)
a	Round-trip amplitude transmission (propagation loss)	dimensionless (0–1)
$T_{through}$, T_{drop}	Through-port and drop-port power transmission spectra	dimensionless (0–1)
FSR	Free spectral range (target/output quantity)	nm
FWHM	Full width at half maximum of the resonance dip/peak	nm
Q_L	Loaded quality factor (target/output quantity)	dimensionless
IL	Insertion loss (target/output quantity)	dB

10.3 Preprocessing and Dataset Notation (Phase 2)

Symbol	Definition	Units
N_{clean}	Final clean sample count after Phase 2.1 filtering	count
$\kappa(g, L_c)$	Engineered feature: analytically-estimated coupling coefficient (Phase 2.5 ablation only)	dimensionless

10.4 Machine-Learning Notation (Phases 3–4)

Symbol	Definition	Units
θ	Trainable weights of the forward network; frozen once reused inside the Tandem Network	—
ϕ	Trainable weights of the inverse network (Phase 4 meaning — see collision note)	—
f_θ	The trained forward network as a function, $f_\theta : D \mapsto R$	—
g_ϕ	The inverse network as a function, $g_\phi : R \mapsto D$	—
F	The true (unknown, deterministic, single-valued but non-injective[v1.3]) geometry-to-response mapping	—
D	A geometry vector, $D = (R, w, g, L_c)$, used generically in Phase 4's derivation	—
D_A, D_B	Two distinct example geometries illustrating the one-to-many problem	—
$\hat{D}$	A geometry predicted by the inverse network, $\hat{D} = g_\phi(R)$	—
R	A response vector, $R = (\lambda_{res}, \mathrm{FSR}, Q_L, \mathrm{IL})$ (Phase 4 meaning — see collision note)	—
N	Number of training samples in a loss summation (Phase 4 meaning — see collision note)	count
$\mathcal{L}_{naive}$	Naive inverse network's parameter-matching loss (Eq. 8)	—
$\mathcal{L}_{tandem}$	Tandem network's response-matching loss (Eq. 9)	—
$\mathcal{L}_{phys}$	Physics-informed regularization loss (Eq. 10, optional)	—
$\mathcal{L}_{total}$	Combined tandem + physics-informed loss (Eq. 11)	—
β	Weight on the physics-informed loss term (Eq. 11, Phase 4 meaning)	dimensionless
$\lambda_{res}^{analytic}(\hat{D})$	Resonance wavelength computed analytically from a predicted geometry	nm
λ_{res}^{target}	The resonance wavelength the inverse query actually asked for	nm

10.5 Evaluation Metric Notation (Phase 6)

Symbol	Definition	Units
n	Number of test-set samples in a metric formula (Eqs. 12–14)	count
x_i^{pred}, x_i^{act}	Predicted and actual value of a target quantity for test sample i	(units of that target)
$\bar{x}^{act}$	Mean of the actual values over the test set	(units of that target)
MAPE, RMSE	Mean Absolute Percentage Error, Root Mean Squared Error	%, (target units)
R^2	Coefficient of determination (Eq. 14) (not ring radius — see collision note)	dimensionless (0–1)
RMS-RGE	RMS Relative Geometric Error, aggregated over $p \in \{R, w, g, L_c\}$ (Eq. 15)	%
p^{pred}, p^{target}	Predicted and target value of one geometric parameter inside RMS-RGE	(units of that parameter)
s_p	Normalizer for parameter p in Eq. 15: p^{target} for R, w, g; the range width ($3\,\mu$m) for L_c[v1.3]	(units of that parameter)

10.6 CPO Application-Mapping Notation (Phase 7)

Symbol	Definition	Units
c	Speed of light in vacuum	m/s
Δf_{3dB}	3-dB optical bandwidth of the resonance (Eq. 16)	Hz
$\Delta\lambda_{FWHM}$	Same quantity as FWHM, written with Δ for the bandwidth formula	nm
N_{ch}	Approximate number of usable WDM channels within one ring's FSR	count
$\Delta\lambda_{ch}$	Target channel spacing for a given WDM grid	nm
IL_{total}, IL_{avg}	Total and average insertion loss across a cascade of rings	dB
N	Number of rings cascaded in a WDM chain (Phase 7 meaning — see collision note)	count
dn/dT	Thermo-optic coefficient of the waveguide core material	K^{-1}
$d\lambda_{res}/dT$, dn_{eff}/dT	Resonance-wavelength and effective-index sensitivity to temperature	nm/K, K^{-1}
ΔT	Temperature change being compensated for	K

Symbol-collision disambiguation — read this before citing an equation out of context: Four symbols carry two distinct meanings across this document, both retained deliberately because each is the standard notation in its own literature (photonics vs. machine learning). Context always disambiguates:

- R means *ring radius* everywhere in Phase 1 and Phase 5, but means a generic *response vector* inside the Phase 4 loss equations, and appears a third way inside R^2 (Phase 6), the unrelated statistical coefficient of determination.
- ϕ means the *round-trip phase* (Eq. 1) in Phase 1, but means the *trainable weights of the inverse network* in Phase 4.
- β means the *propagation constant* (Eq. 1) in Phase 1, but means the *scalar weight on the physics-informed loss term* (Eq. 11) in Phase 4.
- N always means a count of things, but the thing counted changes: training samples (Phase 4) or cascaded rings (Phase 7). Phase 6 deliberately uses lowercase n for the test-set count to stay visually distinct.

When writing the actual thesis text, add a parenthetical reminder at first use in each chapter (e.g. "ϕ, the inverse network's weights") to spare the reader from tracking this across chapters.

11 Background Theory Primer

11.1 What a Microring Resonator Is

A microring resonator is, at its simplest, a closed loop of optical waveguide — a channel, only a few hundred nanometres wide, etched into a thin layer of a transparent material such as silicon and bent around into a circle a few micrometres to a few tens of micrometres in radius. Light is guided along this loop by total internal reflection, in exactly the same physical principle that keeps light trapped inside an optical fibre: the core material (silicon) has a higher refractive index than the material surrounding it (silicon dioxide), so light travelling inside the core reflects off the core–cladding boundary rather than escaping into the cladding.

What makes this simple ring interesting is what happens when a second waveguide — a straight "bus" waveguide carrying the signal that a designer actually wants to route — is placed

close enough to the ring that light can leak across the gap between them. This is not a physical connection; no light-guiding material bridges the gap. Instead, the guided mode's electromagnetic field extends a small distance beyond the waveguide's physical boundary (the "evanescent tail" of the mode), and if a second waveguide sits within that tail, energy can transfer between the two guides purely through the overlap of their fields. This mechanism, evanescent coupling, is the same physical effect that underlies directional couplers, and it is the only way light gets into or out of the ring at all — the ring itself has no input or output port of its own.

Once light has coupled into the ring, it circulates: it travels once around the loop, arrives back at the coupling region, and either couples back out into the bus waveguide or continues around for another lap. This repeated circulation is what makes the structure a resonator. For most wavelengths, light that has gone once around the ring returns to the coupling point slightly out of phase with the light still arriving fresh from the bus waveguide, and the two contributions partially cancel through destructive interference. But at specific wavelengths — those for which the round-trip distance around the ring is an exact whole-number multiple of the wavelength inside the waveguide — the light that has circulated returns perfectly in step with newly arriving light, and the two add constructively. At these special wavelengths, called resonances, optical power builds up inside the ring to levels far higher than what is present in the bus waveguide itself, and the resonator's response to the outside world changes dramatically: a wavelength on resonance can be almost completely diverted out of the through port and into a separate drop port, while a wavelength off resonance passes straight through the first bus waveguide largely undisturbed. This is the physical basis of every application discussed in this thesis: the microring resonator is, functionally, a wavelength-selective switch with no moving parts.

The resonance condition can be written precisely. If n_{eff} is the effective refractive index the guided mode experiences and L is the physical round-trip length of the ring, resonance occurs whenever

$$m\lambda_{res} = n_{eff}L, \qquad m = 1, 2, 3, \ldots$$

for some integer m (the resonance order). This is precisely the phase condition encoded in Eq. 1, where the round-trip phase $\phi = \beta L$ is required to be a multiple of 2π for constructive interference; substituting $\beta = 2\pi n_{eff}/\lambda$ recovers the resonance condition above. Because L depends on both the ring's radius and, in the racetrack geometry used in this thesis, the coupling length L_c (Eq. 1: $L = 2\pi R + 2L_c$), every one of the four free geometric parameters in this thesis's design space ultimately acts through this single resonance condition, either directly (by changing L) or indirectly (by changing n_{eff} through the waveguide's cross-sectional dimensions).

11.2 Coupled-Mode Theory: Where the Add-Drop Equations Come From

The resonance condition above explains where the resonances sit, but not how strong or how narrow they are, nor how much power actually reaches the drop port. Answering those questions requires a slightly more detailed model of what happens at each coupling region, and this is the role of coupled-mode theory (CMT).

The essential idea of CMT, in the time-domain formulation originally developed for exactly this class of device by Little, Chu, Haus, Foresi, and Laine (1997) [14], is to treat the ring not as a detailed electromagnetic structure but as a single abstract "mode" holding some amount of stored optical energy, whose amplitude decays over time for two independent reasons: energy leaking back out through each coupler, and energy being lost to absorption or scattering as it travels around the ring. Each coupler is described by just two numbers: a self-coupling coefficient t, representing the fraction of the field amplitude that passes straight through the coupler without transferring to the other waveguide, and a cross-coupling coefficient κ, representing the fraction that does transfer. For an idealized, lossless coupler, these two quantities are linked by $\kappa^2 + t^2 = 1$: whatever fraction of the light does not stay in its original waveguide must appear in the other one. Propagation loss around the ring itself is captured by a single round-trip amplitude transmission factor a, with $a = 1$ corresponding to a lossless ring and $a < 1$ representing some fraction of the

circulating power dissipated on every lap.

With these three building blocks — t_1 and t_2 at the two couplers, and a for round-trip loss — the frequency response of the whole device can be assembled by adding up, for every possible number of round trips a photon might make before leaving, that path's amplitude and phase, and letting the (infinite) series sum itself into a closed form. Carrying out this sum for a two-bus add-drop ring produces exactly the through-port and drop-port transmission functions given as Eq. 2 and Eq. 3. The important conceptual takeaway is that both equations are nothing more than "how much of an infinite sum of round trips constructively interferes at this wavelength," and that every geometric parameter in this thesis's design space enters that sum only through t_1, t_2, a, and the round-trip phase ϕ. This is precisely why the four design parameters R, w, g, and L_c are sufficient to determine the entire optical response: R and L_c set the round-trip length (and hence the phase ϕ and the free spectral range), w sets the effective index n_{eff} (and hence, together with L, the exact resonance wavelengths), and g together with L_c sets how strongly each coupler transfers light (the values of t_1 and t_2).

[v1.3] **Why the same four parameters also make one-to-many ambiguity plausible here.** The paragraph above is also, read the other way, the clearest statement of the mechanism Objective O6 tests: because L_c enters *both* the round-trip length and the coupler strength, and because g enters only the coupler strength, there exist (g, L_c) pairs that produce nearly the same (t_1, t_2) while differing in L — and compensating R adjustments can restore L. That is a concrete, physically motivated route to $F(D_A) \approx F(D_B)$ in this specific design space, and it is worth stating explicitly in the thesis's Discussion chapter whichever way O6 resolves.

11.3 What Q-Factor, FSR, and Insertion Loss Physically Mean

Three of this thesis's four optical target quantities describe different aspects of the resonance just derived, and it is worth being precise about what each one physically represents.

Free spectral range (FSR) is the wavelength spacing between one resonance and the next. Because the resonance condition $m\lambda_{res} = n_{eff}L$ must hold for consecutive integers m, resonances repeat periodically, and the spacing between them, given by Eq. 4, is set almost entirely by the round-trip length: a physically larger ring supports more wavelengths' worth of round-trip length, so its resonances are packed more closely together (smaller FSR), while a smaller ring has fewer, more widely spaced resonances (larger FSR). This is why ring radius is the dominant lever for controlling how many independent wavelength channels a WDM system built from these rings can support without a resonance accidentally falling on top of the wrong channel.

Quality factor (Q) describes how sharply defined a resonance is, or equivalently, how many round trips a photon typically survives inside the ring before it is lost or before the phase relationship that sustains constructive interference is disturbed. A high-Q resonator holds onto its circulating energy for many round trips, which manifests as a very narrow resonance dip in wavelength (a small FWHM) and, by Eq. 5, a large $Q_L = \lambda_{res}/\text{FWHM}$. A low-$Q$ resonator loses its energy quickly, producing a broad, shallow resonance feature. It is important to note that Q is not an unambiguous "goodness" score to be maximized in every application: a very high Q is exactly what a frequency-comb or squeezed-light source wants (long photon lifetimes enable strong nonlinear buildup), but it is usually undesirable for a data-communication filter, where a very narrow resonance line is also a very narrow usable bandwidth, and becomes impractically sensitive to the small, unavoidable resonance shifts caused by temperature fluctuations or fabrication tolerances. This distinction is the physical basis for the moderate-Q design target adopted throughout this thesis, in explicit contrast to the ultra-high-Q designs pursued elsewhere in the microring literature.

Insertion loss (IL) describes how much of the input optical power fails to reach its intended destination port even under ideal, on-resonance conditions. In an add-drop filter, perfect, lossless operation would mean that 100% of the power at a resonant wavelength reaches the drop port;

insertion loss, defined in Eq. 7 as $-10\log_{10}(T_{drop,peak})$, quantifies the shortfall from that ideal in decibels. Insertion loss has two physical origins that this thesis's design space must jointly manage: intrinsic propagation loss inside the ring itself (captured by a), and coupling mismatch, meaning that the two couplers are not perfectly balanced for the amount of intrinsic loss present. This second cause is why insertion loss cannot be treated as a simple by-product of Q; a ring can be simultaneously high-Q and high-loss (if it is badly undercoupled relative to its intrinsic loss) or comparatively low-Q and low-loss (if it is well matched), which is precisely why this thesis treats insertion loss as an independent target quantity to be predicted and optimized, rather than something that can be inferred for free once Q is known.

11.4 Why Co-Packaged Optics Needs Wavelength-Division Multiplexing

The motivation for building microring filters at all comes from a specific and increasingly urgent bottleneck in modern computing hardware. Data centres and high-performance computing systems move ever-larger volumes of data between processors, memory, and switches, and for the last several decades that movement has relied on electrical signalling over copper traces. Electrical interconnects, however, face a hard physical ceiling: as data rates increase, the electrical loss and signal distortion incurred over even a few centimetres of copper trace increase sharply, and the power needed to drive a signal reliably across that distance grows even faster. Traditionally, this has been solved by placing optical transceivers at the faceplate of a switch or server, connected to the actual switching silicon by a relatively long run of copper trace or cable. Co-packaged optics (CPO) is the industry's response to the fact that even this arrangement is running out of headroom: it means physically integrating the optical transceiver into the same package as the switching or compute silicon, shortening the electrical path to a few millimetres.

This architectural shift only pays off if the optical side of the system can carry enough aggregate bandwidth, and a single wavelength of light on a single fibre or waveguide has a practical bandwidth ceiling of its own. The standard way to multiply the usable bandwidth of a single physical fibre or waveguide is wavelength-division multiplexing (WDM): many independent data channels, each riding on its own, closely-spaced wavelength of light, are combined onto one waveguide at the transmitter and separated back out again at the receiver. This is precisely where the microring resonator earns its place in a CPO system. Because a ring resonator responds selectively to light near its resonance wavelength and passes everything else largely undisturbed, an array of rings, each tuned to a different WDM wavelength, forms a compact, entirely passive (or actively tunable) wavelength multiplexer or demultiplexer, occupying a footprint of only tens of micrometres per channel — orders of magnitude smaller than the alternative technologies that perform the same function at larger scale. The specific numbers this thesis targets — FSR wide enough to host several WDM channels, Q high enough to isolate one channel from its neighbours but not so high that it restricts usable bandwidth, and IL low enough that cascading several such filters does not exhaust the optical power budget — are direct proxies for channel count, channel isolation, and link power budget: the three quantities that determine whether a microring-based WDM filter bank is actually viable inside a real CPO transceiver.

11.5 Why Machine Learning Is Used for Inverse Design

Given the physical model above, one can always compute the optical response of a specific, fully-specified geometry: choose R, w, g, and L_c, run a full-wave electromagnetic simulation, and read off λ_{res}, FSR, Q_L, and IL. This is the forward problem, and it is easy in the relevant sense: it has exactly one answer for any given geometry.

The problem a device engineer actually wants solved usually runs in the opposite direction: given a target set of specifications, what geometry achieves it? This is the inverse problem, and it is harder for two related reasons. First, there is generally no closed-form way to invert the coupled-mode-theory equations directly. Second, and more subtly, the mapping from geometry to response is not guaranteed to be injective: two distinct geometries can produce nearly identical

optical responses, so the correct inverse answer to a given target is not even uniquely defined. Conventionally, the inverse problem is solved by iterative optimization, but every iteration costs a full simulation.

Machine learning offers a different trade: instead of simulating a fresh geometry every time a new target specification arises, a model is trained once, on a fixed dataset of geometry–response pairs generated in advance, to directly approximate either the forward or inverse mapping. Once trained, a query costs a single, near-instantaneous function evaluation rather than a fresh simulation. This thesis benchmarks four ways of learning that mapping: two classical, tree-based ensemble methods (Random Forest and XGBoost), a deep neural network, and a tandem network, the specific architectural refinement designed to address the one-response-many-geometries ambiguity described above — explained in full mathematical detail, including the underlying loss functions, in Phase 4 (Eqs. 8–9). The remainder of this thesis is concerned with putting these four methods through an identical, controlled comparison, with every target quantity understood not as an abstract regression output but as the specific physical proxy for CPO system performance explained above.

12 Related Work Chapter

This chapter is organized thematically rather than paper-by-paper, since several of the twelve reviewed works address the same methodological question from different angles. Citation numbers match the bibliography in Section 13.

12.1 The Tandem Network and the One-to-Many Inverse Design Problem

The starting point for any discussion of machine-learning-based inverse design in photonics is the observation, formalized by Liu et al. [3], that a neural network trained to map an optical response directly to a geometry can fail to converge, not because the network lacks capacity, but because the training data itself can be internally inconsistent: two geometries that are genuinely distinct can produce optical responses so similar that a network sees, in effect, contradictory training instructions for the same input. Liu et al. demonstrated this failure concretely on a multilayer thin-film design problem, showing that a direct inverse network's training loss barely decreased even after fifteen thousand epochs, and that the obvious remedy of filtering out near-duplicate response pairs from the training set only removed the problem partially (their filtered set retained 26.1% of instances and trained no better). Their proposed solution, the tandem network, has since become the standard reference architecture for neural inverse design whenever this ambiguity is a concern, and it is the architecture this thesis benchmarks against classical machine learning and a plain deep neural network.

Two subsequent works situate this result within the broader landscape of deep-learning-based nanophotonic design without extending it to microring resonators specifically. Wiecha et al. [1] provide a critical review of the field's inverse-design strategies at large. Their taxonomy presents conditional GANs, conditional variational autoencoders, and mixture-density networks as the architectures capable of representing multiple non-unique solutions — from which it follows that a converged tandem network recovers only one of what may be several valid solutions to an ambiguous target, a limitation this thesis's own experiments (Objective O6) test directly for the specific case of a four-parameter microring design space.[v1.3] Khaireh-Walieh et al. [2], writing a practitioner-oriented tutorial, state plainly that classical machine-learning methods such as random forests and support vector machines are "computationally more efficient than large deep learning models" for many problems, but explicitly decline to compare them, referring the reader elsewhere on the grounds that this is not the scope of their work. That deferral, made in what is arguably the field's most widely used introductory reference, is direct textual evidence that the classical-ML-versus-deep-learning comparison this thesis performs has not been carried out even at the level of general nanophotonic inverse design.

12.2 Classical Machine Learning for Microresonator Design

Set against that backdrop, Pal et al. [6] is the only work identified in this review that applies classical, tree-based machine learning to microring resonator inverse design at all. Working with a dataset of only 462 finite-element simulations of an isolated silicon-nitride ring, they show that a Random Forest regressor substantially outperforms a single Decision Tree, achieving a mean absolute percentage error at or below roughly 15% across all three predicted parameters when predicting geometry from a ring's integrated dispersion profile, and they validate their trained model against two fabricated, experimentally measured resonators. Their result matters for two reasons beyond its headline accuracy figure. First, it demonstrates that a classical ensemble method can perform credibly on a genuinely small, expensive-to-generate photonic simulation dataset — precisely the data regime this thesis's own Lumerical-based study will operate in. Second, because their design space omits any bus-waveguide coupling parameter entirely, their approach is structurally incapable of predicting insertion loss or any other coupling-dependent quantity — the very specifications that matter most for a co-packaged-optics filtering application. This omission is the most direct and specific gap this thesis addresses.

12.3 Deep Neural Networks for Bus-Coupled Filters and Related Integrated Devices

Three works apply deep neural networks to problems structurally closer to a bus-coupled filter. Chen and Jiang [7] design a four-ring, bus-coupled channel-dropping filter using two independently trained fully-connected networks, generating 25,000 training samples via two-dimensional FDTD simulation in the open-source Meep solver. Notably, their inverse network is not a tandem architecture — it is the same naive direct-inverse design that Liu et al. [3] show can fail — yet Chen and Jiang report good convergence and validate several inverse-designed filters by re-simulating the predicted geometry. This is a genuine point of tension in the literature. [v1.3] Two features of their setup are worth recording precisely, because both bear directly on Objective O6: their design space is unusually *narrow* (the free parameters are radius *offsets* confined to $\pm 0.1\,\mu$m plus three gaps bounded at 0–0.3 μm), and their inverse network receives a *100-point sampled spectrum* as input rather than a handful of scalars. Both properties reduce the likelihood of near-duplicate training responses relative to this thesis's four-scalar formulation. Their work also predicts a full transmission spectrum rather than the compact set of scalar specifications this thesis targets, and it makes no comparison to any classical machine-learning baseline.

Tahersima et al. [4] and Adibnia et al. [9] extend the same forward-plus-inverse deep-learning recipe to two different integrated-photonic device classes. Tahersima et al. design a broadband power splitter on SOI using Lumerical FDTD, generating approximately 20,000 samples, but their device has no resonant behavior at all and is used here chiefly as a calibration point for realistic Lumerical-based dataset scale; their own observation that prediction confidence degrades in the low-transmission regime because the training data skews high-transmission is an early, independent statement of the dataset-coverage problem. Adibnia et al. apply the same architecture to a metal–insulator–metal plasmonic ring resonator, training on 147,456 FDTD-generated samples at a reported data-generation cost of roughly 34 days on three 16-core machines — a concrete, sobering benchmark this thesis uses directly to justify a deliberately smaller, Latin Hypercube-sampled dataset. [v1.3] Their treatment of the many-to-one problem deserves careful reading, because it is frequently cited as evidence that naive inverse networks are fine: they explicitly acknowledge the problem occurring in their own results, note that "the actual structure and the DL-generated structure are distinct", and rest their claim of success on the observation that the two structures' *spectra* agree closely after re-simulation. Their success criterion is therefore a response-space criterion, not a geometry-space one — which is why Objective O6 is stated against both (Section 3). Neither work includes a classical-ML comparison, and neither operates in the linear, low-loss silicon-photonic regime a co-packaged-optics WDM filter requires.

12.4 Hybrid Optimization and Bayesian Approaches

Yeung et al. [5] combine a pretrained generative adversarial network with gradient-based adjoint optimization in a framework named DeepAdjoint. Their central finding — that a target lying in a regime poorly represented by the training dataset leads directly to poor starting designs for the subsequent optimization — is one of several independent confirmations across the reviewed literature that training-data coverage, not model architecture, is often the binding constraint on inverse-design accuracy. Sadeghi Dizaji and Habibiyan [8] apply Bayesian Optimization with a Gaussian-Process surrogate to the design of a microring resonator for quantum squeezed-light generation, achieving strong results from only about one hundred evaluated designs (five sequential rounds of twenty Latin-Hypercube-sampled candidates), and by deriving their parameter bounds directly from analytical Q-factor and free-spectral-range relations before running any simulation at all — a bounds-derivation methodology this thesis's own Phase 1 adopts directly, and the source of the analytical Q_L formula in Eq. 6 (see Fix 1 in the version history for its correction and primary-source cross-check, and Fix 2 in v1.3 for a correction to the substitution described alongside it). Their approach is, however, purpose-built for single-objective sequential optimization toward one target at a time, and their resulting devices sit firmly in the ultra-high-Q regime (Q_L = 1.6–2.8×10^5) that this thesis deliberately avoids.

12.5 Datasets and the Most Recent Advances

Valencia-Garzón et al. [10] published, without applying any machine learning to it at all, an open dataset of 4,719 all-pass microring resonator simulations, validated against two independently published experimental devices to within 2.6–9% relative error on FSR. Their design space omits any coupling-length parameter, but their paper's own User Notes explicitly recommend regression, decision trees, random forest, and neural-network-based inverse design as natural next steps — a direct, published call for exactly the comparison this thesis carries out. Torabi et al. [12], posted as a preprint in 2026, propose a physics-informed graph neural network for the inverse design of a microring resonator biosensor, and define the RMS Relative Geometric Error metric this thesis adopts. Their ablation study — showing that the physics-informed loss term improves test-set generalization even though it does not improve training loss — is directly adapted as an optional extension in Phase 4.

[v1.3] **Fix 11 — corrected characterisation of Paper 12's proximity to this thesis.** Earlier versions described Torabi et al.'s four-parameter design space as "strikingly close" to this thesis's own. It is not. They vary (R, w, h, g) over 0–500 μm, 1.5–3.0 μm, 1.5–3.0 μm and 0.1–0.3 μm — a large polymer-scale biosensor ring, one to two orders of magnitude away from this thesis's SOI telecom filter in three of four dimensions — and they vary waveguide *height* while holding coupling length fixed, which is the exact structural complement of this thesis's choice. The accurate and more useful statement is that Paper 12 is the most recent and most dimensionally comparable microring inverse-design study, and that its continued absence of a classical-ML baseline is the strongest current gap evidence; it is *not* a near-replicate of this thesis's design space. Two further facts should be stated plainly when citing it: it is an unrefereed arXiv preprint, and its dataset comprises 2,500 geometries trained for 80 epochs on an 80/20 split. Its application is refractive-index biosensing, not data communication, and its evaluation includes no classical-ML baseline.

12.6 A Necessary Scoping Note: Microring Resonators as Computing Hardware

Tang et al. [11] use an array of fixed, conventionally-designed microring resonators as the physical weight elements of an optical neuromorphic-computing accelerator. This is the inverse relationship to the one this thesis studies: rather than using machine learning to design a microring resonator's

geometry, Tang et al. use microring resonators as hardware to accelerate someone else's machine-learning computation. No geometry optimization, dataset, or inverse-design procedure of any kind is involved on the photonic side.

Their fabricated device is nonetheless a valuable real-world calibration point, and [v1.3] Fix 12 requires one distinction be drawn that earlier versions did not draw. Their *measured* figures include a 440×220 nm SOI TE single-mode waveguide with 1.3 dB/cm propagation loss, a 20 μm ring radius with a measured FSR of 4.4 nm (implying $n_g = 4.35$, the nominal group index this thesis adopts in Phase 1.2), and drop-port extinction ratios of approximately 30 dB. Their loaded quality factor of $\approx 3 \times 10^5$, by contrast, is **not a measurement** — it appears only as a stated assumption in a forward-looking scalability projection ("we have assumed all MRRs have quality factors of approximately 3×10^5"). It is physically plausible given their propagation loss, but it must be cited as an assumed design value, not as a characterised device parameter. Phase 7 should reference the measured FSR, n_g, and extinction-ratio figures for calibration and treat the Q figure as indicative only.

12.7 Synthesis

Read together, these twelve works trace three separate, only partially overlapping threads: classical machine learning applied to an isolated ring's dispersion [6]; deep neural networks applied to bus-coupled filters or structurally related devices, in every case without a tandem architecture and without a classical-ML comparison [4,7,9]; and the most recent inverse-design architectures, applied to biosensing rather than data communication and still without a classical-ML baseline [12]. The tandem network's founding rationale [3] and its documented limitations [1,2] are well established for nanophotonics in general but have never been tested on a microring resonator specifically. Hybrid optimization methods [5,8] and an openly released, unmodelled dataset [10] sit adjacent to this thesis's core question without answering it. No single study performs a controlled, same-dataset, same-metric comparison of classical ensemble machine learning, a plain deep neural network, and a tandem network, on a bus-coupled add-drop microring resonator with an explicit coupling-length parameter, evaluated against the specific quantities that determine whether such a filter is viable inside a co-packaged-optics interconnect. [v1.1] This specific combination was not identified anywhere in the twelve papers reviewed for this thesis — evidence of a gap in the literature surveyed, not a claim of priority across the field at large (see Section 1.2, Fix 5). The full research-objective statement addressing it is given in Section 3.

13 Formatted Bibliography

IEEE numbered format. References [1]–[12] use the same numbering as the original 12-paper literature review. Reference [13] supports the corrected coupled-mode-theory derivation (Eq. 6, Fix 1); reference [14] is the original time-domain CMT source for the add-drop equations (Eqs. 2–3).

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[12] Y. Torabi, A. Ekhteraei, and M. Khajezadeh, "Physics-informed graph neural network for inverse design of integrated photonic biosensors," arXiv preprint arXiv:2602.19082 [physics.optics], Feb. 2026.

[13] W. Bogaerts, P. De Heyn, T. Van Vaerenbergh, K. De Vos, S. Kumar Selvaraja, T. Claes, P. Dumon, P. Bienstman, D. Van Thourhout, and R. Baets, "Silicon microring resonators," *Laser & Photonics Reviews*, vol. 6, no. 1, pp. 47–73, 2012.

[14] B. E. Little, S. T. Chu, H. A. Haus, J. Foresi, and J.-P. Laine, "Microring resonator channel dropping filters," *Journal of Lightwave Technology*, vol. 15, no. 6, pp. 998–1005, 1997.

Note on reference [12]: Torabi et al. (2026) had not, at the time this document was prepared, appeared in a peer-reviewed venue; cited as an arXiv preprint. Check for an updated peer-reviewed citation before final thesis submission. [v1.3] Its preprint status should also be stated in the thesis text wherever it is described as the most current work, since an unrefereed preprint carries different evidential weight from the refereed papers around it.

Note on reference [7]: No DOI could be independently verified for Chen and Jiang (2022) in earlier versions; [v1.3] the DOI printed on the article itself is `10.1109/JPHOT.2022.3157776` (IEEE Photonics Journal, vol. 14, no. 2, article sequence number 8518911, April 2022). Add it to the bibliography entry.

Note on reference [13][v1.1]**:** Used in Fix 1 to corroborate the corrected Eq. 6. The correction was triangulated via an independent derivation and a direct match against Paper 8's own stated equation (which itself cites [13]); the typeset equation in [13] itself (paywalled, Wiley) was not independently retrieved — confirm directly if you obtain journal access. [v1.3] Eq. 6 was re-derived independently a second time in the v1.3 audit, from the finesse relation, and confirmed; the outstanding item is only the direct reading of [13]'s typeset equation.

14 Uncertainty & Statistical Rigor Protocol

This section formalizes a binding protocol: every accuracy number reported anywhere in the thesis (Phase 3 forward-model accuracy, Phase 4 inverse/tandem accuracy, Phase 5 RF/XGBoost accuracy, and every table in Phase 6) must be produced following the procedure below, not as a single run.

[v1.3] **Framing note (no procedural change).** This protocol is not boilerplate and should not be relegated to an appendix in the thesis write-up. *None* of the twelve reviewed papers reports seed-averaged results with paired significance testing, multiple-comparison correction, and effect sizes; several report a single training run. Stating that plainly in the Methodology chapter is a legitimate, checkable rigor claim and one of the few places where this thesis exceeds rather than matches its comparison set. Present Section 14 as part of the experimental design, not as supplementary material.

14.1 Number of Random Seeds

Fixed default: Train every final model configuration with **5 random seeds**: $\{42, 123, 7, 2024, 31\}$. If compute time is genuinely insufficient, the absolute minimum is 3 seeds ($\{42, 123, 7\}$) — do not report fewer than 3 under any circumstance, and any figure computed from fewer than 3 seeds must be explicitly labelled "single run, not seed-averaged."

What "seed" controls differs by method:

- **DNN and Tandem Network:** the seed controls weight initialization and minibatch shuffling order. The train/validation/test split itself (fixed at seed 42, Phase 2.4) does not change across these 5 runs.
- **Random Forest and XGBoost:** the seed controls `random_state`, governing bootstrap resampling and row/column subsampling. The same fixed split is used for all 5 seeds.

Every hyperparameter search in Phases 3–5 is still run once (against the fixed validation split); only the final, selected configuration is retrained across all 5 seeds.

14.2 Reporting Mean ± Standard Deviation

For any metric x computed once per seed $k = 1, \ldots, K$ (with $K = 5$ by default), report:

$$\bar{x} = \frac{1}{K}\sum_{k=1}^{K} x_k, \qquad s_x = \sqrt{\frac{1}{K-1}\sum_{k=1}^{K}(x_k - \bar{x})^2} \tag{20}$$

using the sample (Bessel-corrected) standard deviation, and present every table entry as $\bar{x} \pm s_x$, never as a bare $\bar{x}$.

14.3 Statistical Test for Comparing Two Methods

When the thesis claims one method is more accurate than another, that claim must be backed by a paired hypothesis test, not by comparing $\bar{x}$ values alone.

Primary test: paired t-test. Because every method is evaluated on the same fixed test set across the same 5 seed indices, the $K = 5$ per-seed scores are naturally paired:

$$d_k = x_k^{(A)} - x_k^{(B)}, \qquad k = 1, \ldots, K \tag{21}$$

$$t = \frac{\bar{d}}{s_d/\sqrt{K}}, \qquad \bar{d} = \frac{1}{K}\sum_{k=1}^{K} d_k, \qquad s_d = \sqrt{\frac{1}{K-1}\sum_{k=1}^{K}(d_k - \bar{d})^2} \tag{22}$$

with $K-1$ degrees of freedom (4, at the default $K=5$), compared against a two-tailed critical value at $\alpha = 0.05$. Use `scipy.stats.ttest_rel` rather than computing this by hand.

Multiple-comparison correction (mandatory when comparing all four methods). A full four-way comparison involves $\binom{4}{2} = 6$ pairwise tests. Apply a Bonferroni correction:

$$\alpha_{corrected} = \frac{\alpha}{6} = \frac{0.05}{6} \approx 0.0083 \tag{23}$$

and report a pairwise difference as statistically significant only if $p < \alpha_{corrected}$.

Robustness check: Wilcoxon signed-rank test. As a robustness check, also run the non-parametric Wilcoxon signed-rank test (`scipy.stats.wilcoxon`) on the same paired differences. Report both results side by side; if they disagree on significance, trust the Wilcoxon result and say so explicitly.

14.4 Effect Size

Report Cohen's d alongside every test:

$$d_{Cohen} = \frac{\bar{d}}{s_d} \tag{24}$$

using the same $\bar{d}$ and s_d from Eq. 22. Interpret using conventional thresholds: $|d_{Cohen}| \approx 0.2$ small, ≈ 0.5 medium, ≥ 0.8 large. A result with a large Cohen's d but $p > \alpha_{corrected}$ should be reported as "a large but not statistically significant difference, likely underpowered at $K = 5$ seeds."

14.5 Worked Reporting Template

Comparison	**Metric**	**Mean ± SD (A)**	**Mean ± SD (B)**	p **(paired** t**)**	**Cohen's** d
RF vs. DNN	MAPE (Q_L)	12.4% ± 1.8%	9.7% ± 2.6%	0.041 ns	1.05

ns denotes "not significant after Bonferroni correction" ($p > 0.0083$).

14.6 Which Metric Gets Tested

Run the full seed/test/effect-size protocol on: (a) the aggregate mean MAPE across the four targets, for the forward-model comparison; (b) RMS-RGE, for the inverse-model comparison; (c) the closed-loop response error, for the final validation claims. Per-target values should still be reported with mean ± SD, but a full pairwise significance test on every individual target for every pair of methods is not required by default; reserve per-target significance testing for any specific claim the thesis text makes about a single target.

15 Future Scope Beyond This Thesis

[v1.3 — NEW SECTION]

Status of this section: entirely outside the scope of this thesis. Nothing in Section 15 modifies, expands, or adds any requirement to Phases 1–7, to Objectives O1–O7, to the Section 5 success criteria, to the Section 7 risk log, or to either schedule in Section 8. It adds no equation, threshold, or deliverable. Every item below is a *follow-up* an MSc or PhD project could pursue, recorded here so that (a) the thesis's Future Work chapter has a ready, evidence-grounded source, and (b) ideas that would otherwise creep into the core plan have a place to live where they cannot do so.

Two items in this section are already flagged as optional/stretch goals inside the core plan (the physics-informed loss in Phase 4.4 step 3, and thermal tunability in Phase 7.5). Their presence here does *not* promote them, and does not change their optional status in those phases. They appear here in the form they would take as full follow-up projects, which is materially larger than the optional version already scoped in Phases 4 and 7.

15.1 Fabrication and Experimental Validation

The single largest limitation of this thesis is stated plainly in the Section 7 risk log: no fabrication access, and therefore an unavoidable simulation-to-reality gap that the Phase 1.6 literature-validation step mitigates but cannot close. A follow-up with foundry access (a multi-project-wafer shuttle run on a standard 220 nm SOI process) could tape out a modest set of the inverse-designed geometries produced in Phase 7 — the Scenario A and Scenario B candidates from each of the four methods — and measure λ_{res}, FSR, Q_L, and IL directly. This is the natural analogue of what Pal et al. [6] did with two fabricated Si_3N_4 rings and what Valencia-Garzón et al. [10] did by validating against two published experimental devices. Two secondary questions become answerable only at this point: how much of the closed-loop error measured in Phase 4.4 survives contact with fabrication tolerance, and whether the per-parameter difficulty ranking from Phase 6.4 (where g is hypothesised to be hardest) mirrors the parameters that fabrication itself controls least precisely.

15.2 Larger and Multi-Fidelity Datasets

This thesis deliberately operates at 800–1,500 samples and measures, rather than assumes, the cost of that choice (Phase 6.3). A follow-up with cluster or cloud compute could extend the dataset toward the scale of Chen & Jiang's 25,000 [7] or Adibnia et al.'s 147,456 [9] samples and test whether the sample-efficiency ranking established in Phase 6.3 inverts at scale — the standard expectation being that tree ensembles lead in the small-data regime and neural methods overtake them given enough data, but this has never been measured for microring inverse design. A cheaper intermediate route is the dual-fidelity strategy already named in the Section 7 risk log and demonstrated by Torabi et al. [12]: pair a large, fast analytical/CMT dataset with a small, expensive FDTD dataset and study the transfer between them explicitly rather than using the analytical set only as an emergency fallback.

15.3 Near-Duplicate-Response Audit of the Design Space

Objective O6 asks whether the one-to-many ambiguity manifests. A natural follow-up asks *why* it does or does not, by characterising the design space directly rather than only observing the models' behaviour in it: compute pairwise distances in the four-dimensional response space across the released dataset and report the fraction of geometry pairs whose responses fall within a measurement-relevant tolerance, as a function of how far apart those geometries are. Liu et al. [3] did the equivalent for thin films via their filtering experiment (26.1% of instances retained, with

little training improvement). No equivalent characterisation exists for a microring design space. This would turn the O6 result from an observation into an explanation, and it would test the (g, L_c) compensation mechanism sketched in Section 11.2 directly. *It is recorded here rather than added to Phase 2 or Phase 6 deliberately, to keep it out of the core plan's critical path.*

15.4 Generalization to Other Device Topologies

The four-parameter symmetric add-drop racetrack studied here is the simplest bus-coupled resonant filter that still contains a genuine coupling-length degree of freedom. Three extensions are natural, and all three are explicitly out of scope above: **(a) asymmetric couplers**, releasing the Phase 1.2 constraint that both couplers share g and L_c, taking the design space from four to six dimensions and allowing the critical-coupling condition to be engineered independently at each port; **(b) higher-order and cascaded filters**, following Chen & Jiang's four-ring channel-dropping topology [7], where coupling-induced frequency shift between rings becomes a first-class design problem that a scalar-target surrogate would have to be re-formulated to represent; and **(c) other resonant topologies** — MZI-assisted rings, Vernier-cascaded rings for FSR extension, and contra-directional couplers — where the question of whether a single surrogate architecture transfers across topologies, or must be retrained per topology, is itself the research question.

15.5 Physics-Informed Losses as a Full Study

Phase 4.4 step 3 scopes a minimal β-ablation of a single resonance-condition penalty term as an optional stretch goal. As a follow-up project this becomes considerably larger: adding penalty terms for the FSR and Q_L relations (Eqs. 4 and 6) as well as resonance, studying the interaction between multiple physics terms, and comparing against the graph-based encoding of device structure that Torabi et al. [12] use to make their network "physics-aware" as distinct from "physics-informed". The specific open question worth asking is whether physics-informed regularization substitutes for training data — i.e. whether it shifts the sample-efficiency curve of Phase 6.3 leftward — which is a different and more useful claim than the generalization improvement Paper 12 reports.

15.6 Thermal Tuning and Control Co-Design

Phase 7.5 scopes a qualitative, optional estimate of $d\lambda_{res}/dT$. A full follow-up would treat thermal behaviour as a design target rather than an afterthought: include heater geometry and tuning power in the design space, add tuning efficiency (pm/mW) and thermal crosstalk between adjacent rings as additional ML targets, and evaluate the resulting designs against a realistic CPO power budget. Tang et al. [11] report $\approx$19.3 mW/π thermo-optic phase-shifter consumption and identify thermal crosstalk as a limiting factor in their own crossbar, which is a concrete starting benchmark. Because resonance drift with temperature is the dominant practical failure mode for microring WDM filters in deployed systems, this is arguably the most industrially relevant of the extensions listed here.

15.7 Fabrication-Tolerance and Yield Modelling

Related to but distinct from Section 15.1: rather than asking what a fabricated device measures, this asks what *distribution* of devices a process yields. Propagating realistic width and gap variation (typically a few nanometres of 3σ lithographic variation) through the trained forward surrogate gives a fast Monte-Carlo estimate of the spec-compliance yield for any candidate geometry — something a surrogate is far better suited to than repeated FDTD. The inverse-design objective could then be restated as "find the geometry with the highest probability of meeting the spec after fabrication" rather than "find the geometry that meets the spec nominally," which is a materially different and more useful optimisation problem. Torabi et al. [12] name fabrication-aware constraints as their own stated next step.

15.8 Multi-Solution Inverse Architectures

Phase 4.2 accepts, as an explicit in-scope limitation, that a converged tandem network returns one valid geometry rather than enumerating all of them. Wiecha et al. [1] catalogue the architectures that lift this restriction: conditional GANs, conditional variational autoencoders, and mixture-density networks, the last of which represents the solution set as a weighted mixture of Gaussians. If Objective O6 finds that ambiguity *does* manifest in this design space, enumerating the full solution set becomes directly valuable rather than academic — a designer could then choose among several valid geometries on secondary criteria (footprint, fabrication tolerance, thermal sensitivity) that were never part of the original target specification. Paper 1 notes the main drawback of the mixture-density approach, namely that the number of non-unique solutions must be approximately known in advance, which is itself a question the Section 15.3 audit would help answer.

15.9 Active Learning and Adaptive Sampling

Phase 1 uses one-shot Latin Hypercube Sampling with a fixed seed, and Phase 2.2 allows one supplementary batch if a coverage gap is found. A follow-up could replace this with a closed-loop active-learning pipeline: train on an initial batch, identify the regions of design space where the surrogate is least certain or where the inverse model performs worst, and spend the next simulation budget there. Yeung et al. [5] propose exactly this as their own future work — "automated feedback loops… which identify areas of weak training data coverage" — and Sadeghi Dizaji & Habibiyan [8] demonstrate the sequential version of the idea with Bayesian Optimization over five batches of twenty designs. Combining the two would test whether an adaptively sampled dataset of N points outperforms a Latin-Hypercube dataset of the same N, which is the practically important question for anyone repeating this work under a compute constraint.

15.10 Benchmark Standardization

Phase 6 exists partly because cross-paper comparison in this field is currently unreliable — Paper 12's own comparison table places a spectral-accuracy percentage, a Chamfer distance, a dispersion improvement over particle-swarm optimisation, an RMSE borrowed from a 4-D flow-MRI study, and an RMS-RGE side by side. A longer-term contribution, well beyond a BSc thesis but directly enabled by the dataset this one releases, would be a standing public benchmark for microring inverse design: a fixed dataset, a fixed split, a fixed metric set, and a leaderboard. The Phase 1.5 step 8 GitHub release is the first component of such a benchmark, and framing it that way in the thesis's Future Work chapter costs nothing and raises the value of the released artifact.

Version History

Version	Date	Change
v1.0	—	Original *Methodology & Reference Plan* (9 sections) and *Supporting Chapters & Reference Appendix* (5 sections) issued as two separate documents.
v1.1	—	Merged into a single source of truth. Five corrections applied, per external audit and *Errata & Patch v1.1*: **Fix 1:** Corrected Eq. 6 (analytical Q_L formula) — was missing a factor of π and silently dropped the coupling-length term; re-derived from first principles and cross-checked against Paper 8's own stated equation. **Fix 2:** Added a Risk Log entry and tool-substitution notes for the case of no Lumerical access at all (MEEP/Tidy3D contingency). **Fix 3:** Added a mandatory timing-calibration sub-study (new Phase 1.6 item 1) with an explicit decision rule, plus a 2-D-FDTD-downgrade fallback in the Risk Log. **Fix 4:** Added an explicit part-time (15–20 hrs/week, ≈48-week) schedule alongside the original full-time (35 hrs/week, 24-week) timeline, with the full-time assumption now stated explicitly. **Fix 5:** Softened novelty framing in the Abstract and Section 2.3 to describe this work accurately as a controlled benchmarking study rather than implying a new method or mechanism.
v1.2	—	Closes two objective-traceability gaps identified by a follow-up hostile audit, resolves one reported (non-)issue, and applies two framing-only improvements. No new sections, phases, or objectives added: **Fix 1 (O5 traceability):** Phase 6.3 now mandates an explicit written verdict — which method is most accurate in the ≤ 300-sample regime and which at full dataset size — after the sample-efficiency curve is produced, matching the reporting-discipline pattern already used in Phase 6.4 and Phase 7.4. **Fix 2 (O1 traceability):** Phase 1.5 now includes an explicit numbered step to push the cleaned dataset, quality log, and validation/convergence reports to a public GitHub repository. (*This entry originally cited "step 9"; corrected to step 8 in v1.3 — see v1.3 Fix 5.*) **Fix 3 (TOC numbering check):** Audited the compiled PDF directly (not extracted/pasted text): Section 6 (Tools & Software Stack) is correctly numbered on the actual page. The "16" previously reported was confirmed to be a chat-paste text-extraction artifact, not a LaTeX section-counter bug. No change made to the document. **Fix 4 (novelty framing, cont.):** Added a framing note in Section 3 (and one sentence in the Abstract, Section 1.2) identifying Objective O6 as this thesis's primary citable scientific contribution, with both possible outcomes (H_A/H_B) stated explicitly as equally valid findings. Added one sentence to Phase 6.3 framing the sample-efficiency curve as this thesis's direct, evidence-based answer to the small-dataset concern a Q1/Q2 reviewer would raise.

Version	Date	Change
v1.3 (this document)	—	**Third independent audit.** Conducted without reliance on the v1.1/v1.2 findings: every numbered equation (Eqs. 1–24) was re-derived from first principles or checked against the paper it is attributed to, and every numeric claim attributed to a source was verified against that source's actual text. **Equations 1–24 are all confirmed correct as typeset**, including the v1.1 Fix 1 correction to Eq. 6, which was independently re-derived and holds. Sixteen corrections are applied below. They are prose-, attribution-, definition-, and framing-level only: *no fixed default, equation, threshold, objective, phase, sub-study, or timeline entry was removed, weakened, or relaxed, and no new phase or objective was added.* One new section (Section 15) was added and is explicitly and entirely outside this thesis's scope. *Substantive corrections (mathematical or definitional):* **Fix 1 (Phase 4.1):** The mapping F was described as "deterministic, one-to-one" in the same sentence that asserted $F(D_A) \approx F(D_B)$ for $D_A \neq D_B$ — a self-contradiction, in the opening line of the derivation underpinning Objective O6. Corrected to *deterministic and single-valued but not injective* (many-to-one), matching the terminology of Papers 3 and 9. **Fix 2 (Phase 1.3):** The note justifying Eq. 6 described the single-coupler-to-two-coupler generalization as $t \to \sqrt{t_1 t_2}$. This is wrong and inconsistent with Eqs. 2–3, which use the product $t_1 t_2$; corrected to $t \to t_1 t_2$, which is what actually reproduces Eq. 6 from Paper 8's Eq. (6). *Eq. 6 itself is unchanged and confirmed.* **Fix 4 (Eq. 15, §10.5, O3):** RMS-RGE now carries an explicit 100%× factor (it previously returned a fraction while being reported in % and thresholded at $\leq 15\%$), and a normalizer convention s_p is fixed, because $L_c = 0$ is a valid design point at which a target-normalised relative error is undefined. Per-parameter reporting and a Paper-12-comparable $\{R, w, g\}$-only value are now required for transparency. *O3's $\leq 15\%$ threshold is unchanged.* **Fix 5 (§6, and the v1.2 entry above):** "Phase 1.5 step 9" corrected to **step 8** in both places; Phase 1.5 has eight steps. **Fix 6 (Phase 7.2/7.3):** Eq. 18 is now explicitly labelled a worst-case cascade bound, with the physical per-channel demux loss (≈1.7 dB for an 8-ring chain, not 8 dB) stated for contrast. *The 1.0 dB/ring Scenario B threshold is retained unchanged*, as conservatism is appropriate for a design-time link budget. **Fix 7 (Phase 1.2, 7.3):** A nominal group index is now fixed ($n_g \approx 4.2$), without which the "FSR ≈ 6–18 nm" rationale could not be checked; corroborated by Paper 11's measured $n_g = 4.35$ for a 440×220 nm SOI TE ring. Paper 7's FSR figures are reclassified as a footprint precedent only (its implied $n_g \approx 3.6$ reflects a 2-D, TM, 0.3 μm-wide structure), the g-range's numeric justification is reassigned to the pending Phase 1.6 sub-study 4 rather than to Papers 7 and 8, and Scenario A's achievable radius is corrected from "$R \approx 5$–$6\,\mu$m" to $R \approx 5.0$–$5.7\,\mu$m with its training-distribution-boundary implication noted. *No range was changed.* **Fix 8 (§2.3, Phase 1.2, 2.3, 1.6, §7):** An advisory scope note records that the 10^5 upper end of the moderate-Q band may not be reachable within $g \leq 350$ nm, and designates Phase 1.6 sub-study 4 as the sole authority on the achieved ceiling. *The design space, the 10^3–10^5 target band, and the Phase 2.3 $\log_{10}$ transform are all unchanged*; the estimate is analytic and explicitly not a basis for change. *Attribution corrections (claims that did not survive checking against*